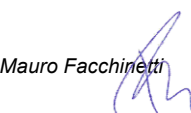
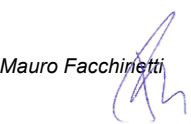
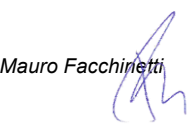
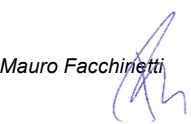


 OFFICINE OROBICHE	RAPPORTO DI COLLAUDO E CERTIFICATI MATERIALI TEST REPORT AND MATERIAL CERTIFICATES (EN 10204 3.1)		DOCUMENTO N° DOCUMENT No.	
			102736 / 111	
CLIENTE: CLIENT:	FLUIDOMATIC		MATRICOLE N° SERIAL NUMBERS:	
ORDINE N° ORDER No.	1.240		F 102736 / 1.1	
RIF. CLIENTE: CLIENT REF.:			SIGLE: TAGS:	
			346LI300	
STRUMENTI: INSTRUMENTS:	INDICATORI DI LIVELLO MAGNETICI MAGNETIC LEVEL GAUGES			
MODELLO: MODEL:	2016.J.2L.G.TL.EX.XX			
QUANTITÀ: QUANTITY:	1			
SOMMARIO CERTIFICATI MATERIALI (VEDERE CERTIFICATI DI ACCIAIERIA ALLEGATI) MATERIAL CERTIFICATES SUMMARY (SEE ATTACHED MILL'S CERTIFICATES)				
Componente / Component		Certif. No. (ME.....)	Componente / Component	
CORPO / BODY		19I00027	VALVOLE DRENO / DRAIN VALVES	
TUBI / PIPES		19F00200		
CHIUSURA / CAP		19L00089	TAPPI / PLUGS	
FLANGE CORPO / BODY FLANGES		19L00195-19I00235	TIRANTI E DADI / BOLTS AND NUTS	
FLANGE DI CONNESSIONE / CONNECTION FLANGES		13592	GALLEGGIANTI / FLOATS	
NIPPLI / NIPPLES		19F00068		
CONTROLLI E COLLAUDI ESEGUITI IN FABBRICA: WORKSHOP TESTS AND CHECKS:				VERIFICHE ISPETTORE: CHECK OF INSPECTOR:
☒ Controllo dei materiali utilizzati Check of employed materials		☐ ☐		☐
☒ Prova in Pressione: Pressure Test:		☒ Idrostatica Hydrostatic	☒ Corpo Body ☒ Galleggiante / Dislocatore Float / Displacer	Pressione di Prova: Testing Pressure: 24 bar g Pressione di Prova: Testing Pressure: 24 bar g Durata: Duration: 5 min
☒ Controllo Visivo e Dimensionale: Visual & Dimensional Check: Connessioni al Processo: Process Connections:		(In accordo a disegno e/o data sheet) (According to drawing and/or data sheet) Lato-lato Side-side Interasse: C-C Distance: 300 mm ☒ Flangiate Flanged 1" 150# RF		Verniciatura: Painting: ☐ Filettate Threaded ☐ A saldare To weld
☐ Prova Rigidità Dielettrica Interruttore di livello/Flusso: Dielectric Strengh Test of Level/Flow Switch:		Vac, 1 min		☐
☐ Misura Resistenza Isolamento Interr. di Livello/Flusso: Insulation Test of Level/Flow Switch:		Vdc > 100 MQ		☐
☐ Controllo Intervento e Differenziale Interruttore di Livello/Flusso (in accordo a data sheet) Check of Switch and Differential of Level/Flow Switch (according to data sheet)				☐
☐ Controllo Portata Flussimetro (in accordo a data sheet) Check of Flowmeter Flow Rate (according to data sheet)				☐
☐ Controllo Taratura Flussimetro: Chekc of Flowmeter Calibration:		(su fondo scala) (on full scale)		☐
☒ Controllo Funzionale Indicatore di Livello Magnetico Functional Test of Magnetic Level Gauge		Visibilità: Visibility: 300 mm		☐
☒ Controllo Funzionale del Trasmettitore 4-20 mA (se presente) Functional Test of 4-20 mA Transmitter (if present)				☐
☐ Controllo Funzionale Indicatore a Lancette (durante assemblaggio parti) Functional Check of Dial Indicator (during assembling of parts)				☐
☐ Controllo Accessori (se presenti): Check of Accessories (if any):				☐
☒ Controllo Targhette Check of Nameplates				☐
☐				☐
RISULTATO CONTROLLO FINALE: FINAL TEST RESULT:				
SODDISFACENTE SATISFACTORY				
Si dichiara che il materiale è stato ispezionato e collaudato in accordo alle procedure interne OFFICINE OROBICHE ed in base al risultato dei controlli sopra indicati risulta conforme ai requisiti dell'ordine, alle specifiche tecniche e i disegni applicabili. We hereby declare that the material has been inspected and tested in accordance with internal procedures OFFICINE OROBICHE and based on result of above mentioned tests it is found to comply with the requirements of the order, with the technical specifications and the applicable drawings.				
Emesso da (Reparto Qualità): Issued by (Quality Department):		Ispettore del Cliente: Client's Inspector:		Ispettore del Cliente: Client's Inspector:
 Mauro Facchinetti				
Data: Date:		Data: Date:		Data: Date:
16/11/19				

 OFFICINE OROBICHE	RAPPORTO DI COLLAUDO E CERTIFICATI MATERIALI TEST REPORT AND MATERIAL CERTIFICATES (EN 10204 3.1)		DOCUMENTO N° DOCUMENT No.	
			102736 / 111	
CLIENTE: CLIENT:	FLUIDOMATIC		MATRICOLE N° SERIAL NUMBERS:	
ORDINE N° ORDER No.	1.240		F 102736 / 2.1	
RIF. CLIENTE: CLIENT REF.:			SIGLE: TAGS:	
STRUMENTI: INSTRUMENTS:	INDICATORI DI LIVELLO MAGNETICI MAGNETIC LEVEL GAUGES		346LI311	
MODELLO: MODEL:	20146.J.2L.G.TL.EX.XX			
QUANTITÀ: QUANTITY:	1			
SOMMARIO CERTIFICATI MATERIALI (VEDERE CERTIFICATI DI ACCIAIERIA ALLEGATI) MATERIAL CERTIFICATES SUMMARY (SEE ATTACHED MILL'S CERTIFICATES)				
Componente / Component		Certif. No. (ME.....)	Componente / Component	
CORPO / BODY		19I00027	VALVOLE DRENO / DRAIN VALVES	
TUBI / PIPES		19F00200		
CHIUSURA / CAP		19L00089	TAPPI / PLUGS	
FLANGE CORPO / BODY FLANGES		19L00195-19I00235	TIRANTI E DADI / BOLTS AND NUTS	
FLANGE DI CONNESSIONE / CONNECTION FLANGES		13592	GALLEGGIANTI / FLOATS	
NIPPLI / NIPPLES		19F00068		
CONTROLLI E COLLAUDI ESEGUITI IN FABBRICA: WORKSHOP TESTS AND CHECKS:				VERIFICHE ISPETTORE: CHECK OF INSPECTOR:
☒ Controllo dei materiali utilizzati <i>Check of employed materials</i>		☐ ☐		☐
☒ Prova in Pressione: <i>Pressure Test:</i>		☒ Idrostatica <i>Hydrostatic</i>		☒
		☒ Corpo <i>Body</i>		
		☒ Galleggiante / Dislocatore <i>Float / Displacer</i>		
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		
		Durata: <i>Duration:</i> 5 min		
☒ Controllo Visivo e Dimensionale: <i>Visual & Dimensional Check:</i>		(In accordo a disegno e/o data sheet) (According to drawing and/or data sheet)		☒
Connessioni al Processo: <i>Process Connections:</i>		Verniciatura: <i>Painting:</i>		
☒ Lato-lato <i>Side-side</i>		☒ Flangiate <i>Flanged</i>		
☒ Interasse: <i>C-C Distance:</i> 300 mm		☒ Filettate <i>Threaded</i>		
		☒ A saldare <i>To weld</i>		
☐ Prova Rigidità Dielettrica Interruttore di livello/Flusso: <i>Dielectric Strength Test of Level/Flow Switch:</i>		Vac, 1 min		☐
☐ Misura Resistenza Isolamento Interr. di Livello/Flusso: <i>Insulation Test of Level/Flow Switch:</i>		Vdc > 100 MQ		☐
☐ Controllo Intervento e Differenziale Interruttore di Livello/Flusso (in accordo a data sheet) <i>Check of Switch and Differential of Level/Flow Switch (according to data sheet)</i>				☐
☐ Controllo Portata Flussimetro (in accordo a data sheet) <i>Check of Flowmeter Flow Rate (according to data sheet)</i>				☐
☐ Controllo Taratura Flussimetro: <i>Check of Flowmeter Calibration:</i>		(su fondo scala) (on full scale)		☐
☒ Controllo Funzionale Indicatore di Livello Magnetico <i>Functional Test of Magnetic Level Gauge</i>		Visibilità: <i>Visibility:</i> 300 mm		☒
☒ Controllo Funzionale del Trasmettitore 4-20 mA (se presente) <i>Functional Test of 4-20 mA Transmitter (if present)</i>				☒
☐ Controllo Funzionale Indicatore a Lancette (durante assemblaggio parti) <i>Functional Check of Dial Indicator (during assembling of parts)</i>				☐
☐ Controllo Accessori (se presenti): <i>Check of Accessories (if any):</i>				☐
☒ Controllo Targhette <i>Check of Nameplates</i>				☒
☐				☐
RISULTATO CONTROLLO FINALE: FINAL TEST RESULT:				
SODDISFACENTE SATISFACTORY				
Si dichiara che il materiale è stato ispezionato e collaudato in accordo alle procedure interne OFFICINE OROBICHE ed in base al risultato dei controlli sopra indicati risulta conforme ai requisiti dell'ordine, alle specifiche tecniche e i disegni applicabili. We hereby declare that the material has been inspected and tested in accordance with internal procedures OFFICINE OROBICHE and based on result of above mentioned tests it is found to comply with the requirements of the order, with the technical specifications and the applicable drawings.				
Emesso da (Reparto Qualità): Issued by (Quality Department):		Ispettore del Cliente: Client's Inspector:		Ispettore del Cliente: Client's Inspector:
				
Data: Date:		Data: Date:		Data: Date:
16/11/19				

 OFFICINE OROBICHE	RAPPORTO DI COLLAUDO E CERTIFICATI MATERIALI TEST REPORT AND MATERIAL CERTIFICATES (EN 10204 3.1)		DOCUMENTO N° DOCUMENT No.	
			102736 / 111	
CLIENTE: CLIENT:	FLUIDOMATIC		MATRICOLE N° SERIAL NUMBERS:	
ORDINE N° ORDER No.	1.240		F 102736 / 3.1	
RIF. CLIENTE: CLIENT REF.:			SIGLE: TAGS:	
STRUMENTI: INSTRUMENTS:	INDICATORI DI LIVELLO MAGNETICI MAGNETIC LEVEL GAUGES		346LI300	
MODELLO: MODEL:	2016.J.2L.G.TL.EX.XX			
QUANTITÀ: QUANTITY:	1			
SOMMARIO CERTIFICATI MATERIALI (VEDERE CERTIFICATI DI ACCIAIERIA ALLEGATI) MATERIAL CERTIFICATES SUMMARY (SEE ATTACHED MILL'S CERTIFICATES)				
Componente / Component		Certif. No. (ME.....)	Componente / Component	
CORPO / BODY		19I00027	VALVOLE DRENO / DRAIN VALVES	
TUBI / PIPES		19F00200		
CHIUSURA / CAP		19L00089	TAPPI / PLUGS	
FLANGE CORPO / BODY FLANGES		19L00195-19I00235	TIRANTI E DADI / BOLTS AND NUTS	
FLANGE DI CONNESSIONE / CONNECTION FLANGES		13592	GALLEGGIANTI / FLOATS	
NIPPLI / NIPPLES		19F00068		
CONTROLLI E COLLAUDI ESEGUITI IN FABBRICA: WORKSHOP TESTS AND CHECKS:				VERIFICHE ISPETTORE: CHECK OF INSPECTOR:
☒ Controllo dei materiali utilizzati <i>Check of employed materials</i>		☐ ☐		☐
☒ Prova in Pressione: <i>Pressure Test:</i>		☒ Idrostatica <i>Hydrostatic</i>		☐
		☒ Corpo <i>Body</i>		☐
		☒ Galleggiante / Dislocatore <i>Float / Displacer</i>		☐
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		Durata: 5 min
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		☐
☒ Controllo Visivo e Dimensionale: <i>Visual & Dimensional Check:</i>		(In accordo a disegno e/o data sheet) (According to drawing and/or data sheet)		☐
Connessioni al Processo: <i>Process Connections:</i>		☒ Lato-lato <i>Side-side</i>		☐
		☒ Flangiate <i>Flanged</i>		☐
		☐ Filettate <i>Threaded</i>		☐
		☐ A saldare <i>To weld</i>		☐
		Interasse: <i>C-C Distance:</i> 300 mm		☐
		1" 150# RF		☐
☐ Prova Rigidità Dielettrica Interruttore di livello/Flusso: <i>Dielectric Strengh Test of Level/Flow Switch:</i>		Vac, 1 min		☐
☐ Misura Resistenza Isolamento Interr. di Livello/Flusso: <i>Insulation Test of Level/Flow Switch:</i>		Vdc > 100 MQ		☐
☐ Controllo Intervento e Differenziale Interruttore di Livello/Flusso (in accordo a data sheet) <i>Check of Switch and Differential of Level/Flow Switch (according to data sheet)</i>				☐
☐ Controllo Portata Flussimetro (in accordo a data sheet) <i>Check of Flowmeter Flow Rate (according to data sheet)</i>				☐
☐ Controllo Taratura Flussimetro: <i>Chek of Flowmeter Calibration:</i>		(su fondo scala) (on full scale)		☐
☒ Controllo Funzionale Indicatore di Livello Magnetico <i>Functional Test of Magnetic Level Gauge</i>		Visibilità: <i>Visibility:</i> 300 mm		☐
☒ Controllo Funzionale del Trasmettitore 4-20 mA (se presente) <i>Functional Test of 4-20 mA Transmitter (if present)</i>				☐
☐ Controllo Funzionale Indicatore a Lancette (durante assemblaggio parti) <i>Functional Check of Dial Indicator (during assembling of parts)</i>				☐
☐ Controllo Accessori (se presenti): <i>Check of Accessories (if any):</i>				☐
☒ Controllo Targhette <i>Check of Nameplates</i>				☐
☐				☐
RISULTATO CONTROLLO FINALE: FINAL TEST RESULT:				
SODDISFACENTE SATISFACTORY				
Si dichiara che il materiale è stato ispezionato e collaudato in accordo alle procedure interne OFFICINE OROBICHE ed in base al risultato dei controlli sopra indicati risulta conforme ai requisiti dell'ordine, alle specifiche tecniche e i disegni applicabili.				
<i>We hereby declare that the material has been inspected and tested in accordance with internal procedures OFFICINE OROBICHE and based on result of above mentioned tests it is found to comply with the requirements of the order, with the technical specifications and the applicable drawings.</i>				
Emesso da (Reparto Qualità): Issued by (Quality Department):		Ispettore del Cliente: Client's Inspector:		Ispettore del Cliente: Client's Inspector:
				
Data: Date:		Data: Date:		Data: Date:
16/11/19				

 OFFICINE OROBICHE	RAPPORTO DI COLLAUDO E CERTIFICATI MATERIALI TEST REPORT AND MATERIAL CERTIFICATES (EN 10204 3.1)		DOCUMENTO N° DOCUMENT No.	
			102736 / 111	
CLIENTE: CLIENT:	FLUIDOMATIC		MATRICOLE N° SERIAL NUMBERS:	
ORDINE N° ORDER No.	1.240		F 102736 / 4.1	
RIF. CLIENTE: CLIENT REF.:			SIGLE: TAGS:	
STRUMENTI: INSTRUMENTS:	INDICATORI DI LIVELLO MAGNETICI MAGNETIC LEVEL GAUGES		346LI311	
MODELLO: MODEL:	2016J.TI.G.TL.EX.XX			
QUANTITÀ: QUANTITY:	1			
SOMMARIO CERTIFICATI MATERIALI (VEDERE CERTIFICATI DI ACCIAIERIA ALLEGATI) MATERIAL CERTIFICATES SUMMARY (SEE ATTACHED MILL'S CERTIFICATES)				
Componente / Component		Certif. No. (ME.....)	Componente / Component	
CORPO / BODY		19I00027	VALVOLE DRENO / DRAIN VALVES	
TUBI / PIPES		19F00200		
CHIUSURA / CAP		19L00089	TAPPI / PLUGS	
FLANGE CORPO / BODY FLANGES		19L00195-19I00235	TIRANTI E DADI / BOLTS AND NUTS	
FLANGE DI CONNESSIONE / CONNECTION FLANGES		13592	GALLEGGIANTI / FLOATS	
NIPPLI / NIPPLES		19F00068		
CONTROLLI E COLLAUDI ESEGUITI IN FABBRICA: WORKSHOP TESTS AND CHECKS:				VERIFICHE ISPETTORE: CHECK OF INSPECTOR:
☒ Controllo dei materiali utilizzati <i>Check of employed materials</i>		☐ ☐		☐
☒ Prova in Pressione: <i>Pressure Test:</i>		☒ Idrostatica <i>Hydrostatic</i>		☐
		☒ Corpo <i>Body</i>		☐
		☒ Galleggiante / Dislocatore <i>Float / Displacer</i>		☐
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		Durata: 5 min
		Pressione di Prova: <i>Testing Pressure:</i> 24 bar g		☐
☒ Controllo Visivo e Dimensionale: <i>Visual & Dimensional Check:</i>		(In accordo a disegno e/o data sheet) (According to drawing and/or data sheet)		☐
Connessioni al Processo: <i>Process Connections:</i>		Verniciatura: <i>Painting:</i>		☐
☒ Lato-lato <i>Side-side</i>		☒ Flangiate <i>Flanged</i>		☐ Filettate <i>Threaded</i>
☒ Interasse: <i>C-C Distance:</i> 300 mm		☐ A saldare <i>To weld</i>		☐
		☐ 1" 150# RF		☐
☐ Prova Rigidità Dielettrica Interruttore di livello/Flusso: <i>Dielectric Strengh Test of Level/Flow Switch:</i>		Vac, 1 min		☐
☐ Misura Resistenza Isolamento Interr. di Livello/Flusso: <i>Insulation Test of Level/Flow Switch:</i>		Vdc > 100 MQ		☐
☐ Controllo Intervento e Differenziale Interruttore di Livello/Flusso (in accordo a data sheet) <i>Check of Switch and Differential of Level/Flow Switch (according to data sheet)</i>				☐
☐ Controllo Portata Flussimetro (in accordo a data sheet) <i>Check of Flowmeter Flow Rate (according to data sheet)</i>				☐
☐ Controllo Taratura Flussimetro: <i>Chek of Flowmeter Calibration:</i>		(su fondo scala) (on full scale)		☐
☒ Controllo Funzionale Indicatore di Livello Magnetico <i>Functional Test of Magnetic Level Gauge</i>		Visibilità: <i>Visibility:</i> 300 mm		☐
☒ Controllo Funzionale del Trasmettitore 4-20 mA (se presente) <i>Functional Test of 4-20 mA Transmitter (if present)</i>				☐
☐ Controllo Funzionale Indicatore a Lancette (durante assemblaggio parti) <i>Functional Check of Dial Indicator (during assembling of parts)</i>				☐
☐ Controllo Accessori (se presenti): <i>Check of Accessories (if any):</i>				☐
☒ Controllo Targhette <i>Check of Nameplates</i>				☐
☐				☐
RISULTATO CONTROLLO FINALE: FINAL TEST RESULT:				
SODDISFACENTE SATISFACTORY				
Si dichiara che il materiale è stato ispezionato e collaudato in accordo alle procedure interne OFFICINE OROBICHE ed in base al risultato dei controlli sopra indicati risulta conforme ai requisiti dell'ordine, alle specifiche tecniche e i disegni applicabili.				
<i>We hereby declare that the material has been inspected and tested in accordance with internal procedures OFFICINE OROBICHE and based on result of above mentioned tests it is found to comply with the requirements of the order, with the technical specifications and the applicable drawings.</i>				
Emesso da (Reparto Qualità): Issued by (Quality Department):		Ispettore del Cliente: Client's Inspector:		Ispettore del Cliente: Client's Inspector:
				
Data: Date:		Data: Date:		Data: Date:
16/11/19				

DUEVI

FLANGES AND FITTINGS

DUEVI SRL
Via Albert Sabin, 15 20873 - CAVENAGO BRIANZA (MB)
Tel. +39 02 95335174 Fax +39 02 95335175
Registro imp. Monza C.F./Partita IVA/VAT no. IT 06473230966
E-mail info@duevifornitureindustriali.it
Web www.duevifornitureindustriali.it

SISTEMA QUALITA'
CERTIFICATO ISO 9001:2015
RINA NR. 7026/02/S

INSPECTI ON CERTI FI CATE EN 10204 : 2004 / 3.1			Nr. / No.	002742	Data / Dated	11/10/2019
OFFI CI NE OROBI CHE SRL VIA SERENA, 10 24010 PONTERRANICA - BG IT			DDT / Delivery note 2671			
			Packing List			
			Pag. 1			
			Vs.Ordine nr./Yr. P.O. no. MB VS ORD. NR. 1709 DEL 17/09/2019			
			Posizione / Item 002			
			Ns. rif. / Our ref.			
Codice colata Heat Code	Nr. colata Heat No.	Origine Origin of Steel	Quantita' Quantity	Descrizione Description		
	901144 120	SIDENOR	200	FL. CORPO F316L x INDICATORE DI LIVELLO		

SERIE 2000 A DIS. OFFICINE OROBICHE SPA
DWG. FL-162 REV. 2

Elementi / Elements	C	Si	Mn	S	P	Cr	Ni	Mo	Ti	Cu	V	Nb	N	Al
LADLE ANALYSIS	0,019	0,484	1,350	0,025	0,030	16,740	10,090	2,030		0,532			0,075	
CE LF														
4,706														
CE LF= C+ Mn/6+ (Cr+ Mo+ V)/5+ (Ni+ Cu)/15														
CE SF= C+ Mn/6														
F1= Cu+ Ni+ Cr+ Mo+ V														
F2= Cr+ Mo														
PREN														
DUREZZA	Snervamento> 0,2%				Snervamento> 1,0%				Rottura	Allungamento	Contrazione			
HARDNESS	Yeld Strength> 0,2%				Yeld Strength> 1,0%				Tensile	Elongation	Reduction of area			
HBW	MPa				MPa				MPa	%	%			
167	270,0								554,0	45,1	56,8			
Provetta	Forma	RESI LI ENZA / I MPACT TEST												
Test specimen	Shape													
Sez/ Sect mm2	L. mm	1=○ - 2=□	°C											
			Tipo/ Type	Provetta/ Test Specimen	°C		1-Joule	2-Joule	3-Joule	Media/ Average				
							0,00	0,00	0,00	0,00				

Materiale in accordo a: ☒ ASTM/ASME CODE SECT. II, PART. A, 2017 EDITION ASTM/ASME A/SA182 F316/316L
Material according to: ☒ NACE MR-0175/2009 ISO 15156
☒ NACE MR-0103/2012
☒ Q.A.S.IN ACCORD.WITH PRESSURE EQUIPMENT DIRECT. 2014/68/UE(PED) ANNEX i, PARAGRAPH 4.3

Trattamento termico Heat treatment	SOLUBILIZZATO Solution treatment and quenched	Produzione Acciaio Steel Production	ELABORATO AL FORNO ELETTRICO Made by electric furnace
Dimensioni in accordo a: Dimensions according to:			
Marcatura in accordo a: Marking according to:		Vis. & Dim.	CONTROLLO VISIVO E DIMENSIONALE SODDISFACENTE Satisfactory visual and dimensional test

(1) (2) (3) (4) Note / Notes Conforme al certificato originale rilasciato dall'acciaieria o forgiatore e relativo laboratorio di analisi e prove.
B D X X According to original certification issued by steel plant and relative analysis and tests laboratory.

Gli articoli sopra certificati sono in accordo alle norme e alle richieste dell'ordine e sono stati prodotti secondo un sistema qualita' approvato e certificato ISO 9001:2015
The certified products comply with specifications and requirements of the order and have been manufactured according to a quality system approved and registered by ISO 9001:2015

UFFI CI O CONTROLLO QUALI TA' QUALI TY CONTROLDEPARTMENT DUEVI s.r.l.	UFFI CI O CONTROLLO QUALI TA' I NSPECTI ON AUTORI THY	MARCHI O PRODUZIONE MANUFACTURER'S SYMBOL
--	--	--



COGNE ACCIAI SPECIALI S.p.a.

ITALY - 11100 AOSTA - VIA PARAVERA, 16
TEL +39.0165.3021 - FAX +39.0165.302296
CAP. SOC. 100.000.000 EUR INT. VERS.
C.F. 02187360967
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with management systems IGO
approved and certified according to ISO 9001,
ISO/TS 16949 e ISO 14001.

INSPECTION CERTIFICATE 3.1 (EN 10204:2004)
DOCUMENT NUMBER 2017047643
PAGE 1/3

CUSTOMER:	PANDA MECCANICA SRL		
CUSTOMER'S ORDER:	mail		
MANUFACTURER'S WORKS:	AOSTA, VIA PARAVERA 16 - ITALY		
PRODUCER OF THE DOC:	QUALITY DEPARTMENT		
MANUFACTURER'S WORKS ORDER NO:	1099684 /90	MARK OF THE MANUFACTURER:	COGNE
DISPATCH NOTE:	81155687 /30	WEIGHT:	140,00
THE CERTIFIED PRODUCTS ARE COMPLYING TO THE PURCHASE ORDER			
INTERNAL SPEC:	STOCKBARREW2	TECHNICAL RULE:	AD2000
PRODUCT:	2257 PEL PEELED	ROUND BARS TOLERANCE:	ISOK11
SURFACE FINISH:	2B Cold Finished		
PRODUCT DELIVERY CONDITION:	RS SOLUTION ANNEALED		
PRODUCT DIMENSIONS (mm):	60,000	LENGTH (mm)	06000 /06200
GRADE:	AISI 316/316L EN1.4401/4404	INTERNAL GRADE:	F316L IMCO
IDENTIFICATION HEAT NO:	772948	IDENTIFICATION LOT NO:	427590
MARKING OF THE PRODUCT:	1.4404	TEST PIECE N:	759

REFERENCE NORMS: EN 10088-3, EN 10272, ASTM A276, ASTM A479, ASTM A193 B8MCLASSE 1D, ASME SA276, ASME SA479, AMS 5653, AMS 5648, ISO15156-3/NACE MR0175, NACE MR0103, NORSOK M630 MDSS01 rev5.
REFERENCE NORMS FOR CHEMICAL COMPOSITION AND MECHANICAL PROPERTIES: ASTM A182, ASME SA182.
REFERENCE NORMS ONLY FOR CHEMICAL COMPOSITION: EN 10222-5, ASTM A484, ASTM A403, ASTM A314, AMS-QQ-S-763.
THE PRODUCT IS CONFORMING TO THE CONTRACT AGREEMENT AND TO THE RULE AD2000-MERKBLATT W0/W2.
ASTM A484 DOES NOT APPLY FOR SIZE AND LENGTH TOLERANCES.
STEELMAKING EAF + AOD + CONTINUOUS CASTING
REFERENCE NORM ASME SEC. II PART A LAST EDITION.
HOT ROLLED
REDUCTION RATIO 8,3

CHEM. COMP. (%W/W) - LADLE ANALYSIS ACCORDING TO ASTM E1019-A751-E1086-E415-A580
Control lot No. - Weight :020000699650 - 86.143 KG

ELEMENTS	C	Si	Mn	P	S	N	Cr	Mo	Ni	Cu
OBTAINED	0,015	0,40	1,84	0,026	0,026	0,084	16,60	2,09	10,13	0,35
ELEMENTS	Co									
OBTAINED	0,17									

HARDNESS TEST IN AS DELIVERY CONDITION
Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION ENISO6506 HARDNESS TEST HB TESTING METHOD: 10/3000
OBTAINED 182

IMPACT TEST IN AS DELIVERY CONDITION
Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION EN ISO 148-1-KV2 DIRECTION OF THE TEST PIECE: L
TYPE OF TEST PIECE KV
TEST TEMPERATURE °C 20
MEASUREMENT UNIT J
OBTAINED 278 279 285 286 280 278 285 290 290 289
280 288



COGNE ACCIAI SPECIALI S.p.a.

ITALY - 11100 AOSTA - VIA PARAVERA, 16
TEL +39.0165.3021 - FAX +39.0165.302296
CAP. SOC. 100.000.000 EUR INT. VERS.
C.F. 02187360967
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with management systems IGQ
approved and certified according to ISO 9001,
ISO/TS 16949 e ISO 14001.

INSPECTION CERTIFICATE 3.1 (EN 10204:2004)
DOCUMENT NUMBER 2017047643
PAGE 2/3

TENSILE TEST IN AS DELIVERY CONDITION

Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION EN-ISO6892-1

DIRECTION OF THE TEST PIECE: L

MEASUREMENT UNIT	RM MPA	RP02 MPA	A5 %	Z %	RP1 NMM
OBTAINED	615	270	5.0 D	80,5	305
	613	264	64,7	83,6	298
	617	275	63,7	81,5	312
	612	276	64,5	83,7	304
			63,2		

GRAIN SIZE

Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION ASTM E112 --
OBTAINED 5 7

IMPACT TEST

Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION EN ISO 148-1
TYPE OF TEST PIECE KV
TEST TEMPERATURE °C 196-
MEASUREMENT UNIT J
OBTAINED 175 178 170 186 181 187 174 171 182 172
190 200

DIRECTION OF THE TEST PIECE: L

TENSILE TEST

Control lot No. - Weight :020000700444 - 24.000 KG
SPECIFICATION ASTM A370-E8-0.5"SPEC

DIRECTION OF THE TEST PIECE: L

MEASUREMENT UNIT	RMB KSI	RP02B KSI	ZB %	AL2B %
OBTAINED	89	38	4,0 D	67,3
	87	38	81,4	67,0
	88	40	82,7	67,3
	88	40	83,3	67,3
			82,2	68,6

COUNTRY OF ORIGIN: ITALY, THE MATERIAL COMPLIES WITH FAR DFARS
252.225-7009 ALT 1.

The products delivered are conforming to EN10088-5 and bear the CE marking on the basis of the test results here reported and the continuous surveillance, assessment and approval by MPA NRW
(certificate no. 0432-CPR-00166-01)

Material produced according to the system #Regulation 305/2011/EU of the European Parliament and of the Council of 9 March 2011#
DOCUMENT ISSUED IN AGREEMENT WITH TUEV BAYERN(11.1972) WAVING OF COUNTERSIGNING (SEE TUEV BAYERN LETTER DATED 17.01.80).

MARKING: SYMBOL OF THE MANUF. WORK, MATERIAL N., CAST N., LOT N., RECEIVING AGENT STAMP

CORROSION TEST ACCORDING TO ASTM A262 Practice E EN ISO 3651-2: SATISFACTORY.

ANTIMIXING TEST BY PORTABLE SPECTROMETRE: CONFORMING



COGNE ACCIAI SPECIALI S.p.a.

ITALY - 11100 AOSTA - VIA PARAVERA, 16
TEL +39.0165.3021 - FAX +39.0165.302296
CAP. SOC. 100.000.000 EUR INT. VERS.
C.F. 02187360967
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with management systems - ISO
approved and certified according to ISO 9001,
ISO/TS 16949 e ISO 14001.

INSPECTION CERTIFICATE 3.1 (EN 10204:2004)
DOCUMENT NUMBER 2017047643
PAGE 3/3

EDDY CURRENT INSPECTION OF SURFACE ACCORDING TO EN 10277-1 CLASS 4.
DIMENSIONAL CONTROL ACCORDING TO EN 10278: SATISFACTORY.
NO WELDING REPAIR ON THE MATERIAL.
ULTRASONIC TEST ACCORDING TO EN 10308 Class.3
INSPECTOR'S STAMP LF

THE PRODUCT SATISFIES CE DIRECTIVES: 2011/65 (ROHS2) - 2000/53 - 2002/95(ROHS) - 2003/11 - 2005/618 AND PED 2014/68/EU.
THE MATERIAL IS FREE OF ANY MERCURY, MERCURY COMPOUNDS AND OR RADIUM CONTAMINATION AT TIME OF SHIPMENT AND WAS PRODUCED WITHOUT
USING OZONE DEPLETING SUBSTANCES OF CLASS I AND II.

CE 0432



Acciaierie Valbruna S.p.A.

36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
Telefono 0444.968211 - Fax 0444.963836
Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
Telefono 0471.924111 - Fax 0471.924497

CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION

In conformità a : EN 10204 (2004) , 3.1 / ISO 10474 (2013) , 3.1
Nach/According to/Selon

Certificato nr. MEST373295 / 2019 /
Prüfung/Test/Essai

Cliente / Besteller/Purchaser/Client
PANDA MECCANICA SRL
VIA IV NOVEMBRE, 5
24050 - GHISALBA - BG

Stato di fornitura : Laminato - Solubilizzato Pelato
Lieferzustand
Delivery state
Etat de livraison

Produttore :
Hersteller/Item/Usine productrice

ACCIAIERIE VALBRUNA S.P.A.

Ordine nr: VS ORDINE MAIL
Bestell
Your order
Commande

Tipo di Elaborazione: E+AOD
Erschmelzungsart
Melting process
Mode d'elaboration

Marchi di Fabbrica:
Zeichen des Lieferwerkes
Trade marks
Sigles de l'usine productrice



Conferma ordine nr: BS19000802
Werks/Our Order/Ref nr.

Qualità: 1.4401/1.4404/316/316L
Werkstoff/Grade/Nuance

Punzone del Collaudatore:
Stempel des Werkssachverständigen
Inspector's stamp/Poinçon de l'assesseur

MR

Avviso di Spedizione: A-BS19000422
Lieferanzeige/Packing list/B.L.

Marca: MVAPML (MAXIVAL)
Markenbezeichnung
Brand / Nuance

Punzonatura: 1.4401/4/316/L
Kennzeichnung
Marking
Marquage

SPECIFICHE :

Note:

Anforderungen / Requirements / Exigences

Aufzeichnungen / Notes / Notes

VAL STOCK 2010 1.4404/316L A
MDS S01 5 316 A (0)
ISI 316
ISI 316L
AMS 5648 M S31600 A
AMS 5653 H S31603 A
ASME SA182 2017 S31600 A (1)
ASME SA182 2017 S31603 A (2)
ASME SA193 2017 B8M CLASS 1 (3)
ASME SA276 2017 S31600 A (4)
ASME SA276 2017 S31603 A (5)
ASME SA320 2017 B8M CLASS1 (6)
ASME SA479 2017 S31600 A (7)
ASME SA479 2017 S31603 A (8)
ASTM A182 2018 S31600 A (9)
ASTM A182 2018 S31603 (A)
ASTM A193 2017 B8M CLASS 1
ASTM A262 2015 PRACTICE E
ASTM A276 2017 S31600 A
ASTM A276 2017 S31603 A
ASTM A314 2015 S31600
ASTM A320 2017A B8M CLASS1
ASTM A370 2017 .
ASTM A479 2018 S31600 A
ASTM A479 2018 S31603 A
ASTM E10 2017 .
ASTM E8 2016A .
DIN 17440 96 1.4401 A
DIN 17440 96 1.4404 A
EN 10088-3 2014 1.4401 A
EN 10088-3 2014 1.4404 A
EN 10272 2007 1.4401 A
EN 10272 2007 1.4404 A
ISO 6506-1 2014 .
ISO 6892-1 2016 .
NACE MR0103 2015 S31600 A (B)
NACE MR0103 2015 S31603 A (C)
NACE MR0175 2015 S31600 A (D)
NACE MR0175 2015 S31603 A (E)
QQ-S-763 F 316 A
QQ-S-763 F 316L A

(0)Norsok-standard M-630 Edition 6, October 2013

(1)Section II Part A 2017 EDITION For products machined directly from bar refer to ASME SA479.
(2)Section II Part A 2017 EDITION For products machined directly from bar refer to ASME SA479.
(3)Section II Part A 2017 EDITION
(4)Section II Part A 2017 EDITION
(5)Section II Part A 2017 EDITION
(6)Section II Part A 2017 EDITION
(7)Section II Part A 2017 EDITION
(8)Section II Part A 2017 EDITION
(9)For products machined directly from bar refer also to ASTM A479.
(A)For products machined directly from bar refer also to ASTM A479.

(B)ANSI/NACE MR0103/ISO 17945 November 23, 2015
(C)ANSI/NACE MR0103/ISO 17945 November 23, 2015
(D)ANSI/NACE MR0175/ISO 15156-3, third edition November 23,2015
(E)ANSI/NACE MR0175/ISO 15156-3, third edition November 23,2015

Vicenza, 27/02/19

VCQ012 - MEST373295

(Mod. MCE2) VGP/910377/06331246/6960C12667C1A06331

Il collaudatore di stabilimento / der Werkssachverständige / Works inspector / L'agent d'usine

M.RIZZOTTO

Pagina - 1 di 3



Acciaierie Valbruna S.p.A.

36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
Telefono 0444.968211 - Fax 0444.963836
Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
Telefono 0471.924111 - Fax 0471.924497

CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION

In conformità a : EN 10204 (2004) , 3.1 / ISO 10474 (2013) , 3.1
Nach/According to/Selon

Certificato nr. MEST373295 / 2019 /
Prüfung/Test/Essai

Cliente / Besteller/Purchaser/Client
PANDA MECCANICA SRL
VIA IV NOVEMBRE, 5
24050 - GHISALBA - BG

Stato di fornitura : Laminato - Solubilizzato Pelato
Lieferzustand
Delivery state
Etat de livraison

Produttore :
Hersteller/Item/Usine productrice

ACCIAIERIE VALBRUNA S.P.A.

Ordine nr: VS ORDINE MAIL
Bestell
Your order
Commande

Tipo di Elaborazione: E+AOD
Erschmelzungsart
Melting process
Mode d'elaboration

Marchi di Fabbrica:
Zeichen des Lieferwerkes
Trade marks
Sigles de l'usine productrice



Conferma ordine nr: BS19000802
Werks/Our Order/Ref nr.

Qualità: 1.4401/1.4404/316/316L
Werkstoff/Grade/Nuance

Punzone del Collaudatore:
Stempel des Werkssachverständigen
Inspector's stamp/Poinçon de l'assesseur

MR

Avviso di Spedizione: A-BS19000422
Lieferanzeige/Packing list/B.L.

Marca: MVAPML (MAXIVAL)
Markenbezeichnung
Brand / Nuance

Punzonatura: 1.4401/4/316/L
Kennzeichnung
Marking
Marquage

		Tolleranza: k12					
		Tolleranza/Allowance/Tolerance					
Pos. nr.	Oggetto	Dimensioni - mm		Lunghezza - mm	Colata	Pezzi	Peso - KG
Pos. nr. Item nr. Nr. de poste	Gegenstand Product description Descrip. du produit	Abmessungen Dimension Dimension		Länge Length Longueur	Schmelze Heat Coulée	Stückzahl Pieces Pièces	Gewicht Weight Poids
0020	Tondo	120,000		5180/ 5180	276866	1	189,0
							Lotto nr.
							Losnr. Lot nr. Lot nr.
							813411310

TEST ALLO STATO DI FORNITURA													
Test on delivery condition Prüfung auf lieferbareitem produkt test a l'etat de fourniture Prueba sobre el material así como entregado													
1) L=longitudinale/längs, T=trasversale/quer, Q=Tangenziale/tangential													
TEST	Provetta/ Probestab Specimen/Eprouvette Larg.diam Spess. Breite Diam. Dicke Width Diam. Thickness Larg. diam. epais mm	°C	Posiz. Saggio Probenlage Location Emplacement 1)	Snervamento Streckgrenze Yield Stress Limite elastique Rp 0,2% N/mm2	Snervamento Streckgrenze Yield Stress Limite elastique Rp 1% N/mm2	Resistenza Zugfestigkeit Tensile strength Resistance à traction Rm N/mm2	Allungamento Bruchdehnung Elongation Reduction of area Striction A5 % E 4d %		Strizione Einschnürung Reduction of area Striction Z % RA %		Resilienza Kerbschlagarbeit Impact Value Resilience KV J		Durezza Haerte Hardness Durete HB
Valori richiesti		min		207	240	517	40	40	-	50	100		140
Anforderungen/Required values Valeurs demandées		max				690							215
A	10	20	L	270	312	581	63	65	68	68	283	283	287
											169		

TEST ALLO STATO DI FORNITURA				
Test on delivery condition / Prüfung auf lieferbareitem produkt / Test a l'etat de fourniture / Prueba sobre el material así como entregado				
TEST				
	min	max		
A	Dimensioni grano x ASTM E112			4

Mechanical properties according to ASTM A370.

Brinnell hardness according to ASTM E10

Tensile testing according to ASTM E8

Brinnell hardness according to ISO 6506-1

Tensile testing according to ISO 6892-1

Analisi chimica													
Chemische Zusammensetzung/Chemical Analysis/Analyse chimique													
Colata /Heat Schmelze/Coulée	min - max	-	1,25 2,00	16,50 18,00	2,00 2,50	-	10,00 13,00	-	0,040	0,030	0,100	-	-
	C %	Si %	Mn %	Cr %	Mo %	Cu %	Ni %	P %	S %	N %			
276866	0,015	0,46	1,46	16,64	2,06	0,54	10,14	0,029	0,026	0,058			

Corrosion test in 16% sulfuric acid and copper sulfate solutions							
Test standard: ASTM A262-Practice E							
Test	Heat treatment before test	Length of Period (h)	Test temp (°C)	Bend Angle (°)	Ø spindle (mm)	Result of visual inspection at 20 x magnification after bend test	Result
Intergranular corrosion	Sensitization	15	boil	180	5	Absence of cracks	SATISFACTORY



Acciaierie Valbruna S.p.A.

36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
Telefono 0444.968211 - Fax 0444.963836
Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
Telefono 0471.924111 - Fax 0471.924497

CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION

In conformità a : EN 10204 (2004) , 3.1 / ISO 10474 (2013) , 3.1

Certificato nr. MEST373295 / 2019 /

Nach/According to/Selon

Prüfung/Test/Essai

Cliente / Besteller/Purchaser/Client
PANDA MECCANICA SRL
VIA IV NOVEMBRE, 5
24050 - GHISALBA - BG

Stato di fornitura : Laminato - Solubilizzato Pelato
Lieferzustand
Delivery state
Etat de livraison

Produttore :
Hersteller/Item/Usine productrice

Ordine nr: VS ORDINE MAIL
Bestell
Your order
Commande

Tipo di Elaborazione: E+AOD
Erschmelzungsart
Melting process
Mode d'elaboration

Marchi di Fabbrica:
Zeichen des Lieferwerkes
Trade marks
Sigles de l'usine productrice

Conferma ordine nr: BS19000802
Werks/Our Order/Ref nr.

Qualità: 1.4401/1.4404/316/316L
Werkstoff/Grade/Nuance

Punzone del Collaudatore:
Stempel des Werkssachverständigen
Inspector's stamp/Poinçon de l'assayeur

Avviso di Spedizione: A-BS19000422
Lieferanzeige/Packing list/B.L.

Marca: MVAPML (MAXIVAL)
Markenbezeichnung
Brand / Nuance

Punzonatura: 1.4401/4/316/L
Kennzeichnung
Marking
Marquage

Corrosion test in 16% sulfuric acid and copper sulfate solutions

Test standard: UNI EN ISO 3651-2 Method A

Test	Heat treatment before test	Length of Period (h)	Test temp (°C)	Bend Angle (°)	Ø spindle (mm)	Result of visual inspection at 20 x magnification after bend test	Result
Intergranular corrosion	Sensitized T1	20	boil	90	5	Absence of cracks	SATISFACTORY

Reduction ratio = 4,1 : 1
Solution annealed 1070° C for 2.5h/water.

Sono state soddisfatte tutte le condizioni richieste
Die gestellten Anforderungen sind it. Anlage erfüllt
The material has been furnished in accordance with the requirements
Le matériel a été trouvé conforme aux exigences

Controllo antimescolanza: OK
Verwechslungsprüfung: spectralanalytisch durchgeführt
Antimixing testing performed: OK
Contrôle antimélange fait: r.a.s.

Controllo visivo e dimensionale: soddisfa le esigenze
Besichtigung und Ausmessung: ohne Beanstandung
Visual inspection and dimensional checks:satisfactory
Contrôle visuel et dimensions: satisfaisant

Melted and manufactured in Italy

No welding or weld repair

Material free from Mercury contamination

We declare that the finished product is checked for radioactive contamination through Portal System when it leaves the production plant.

QUALITY MANAGEMENT SYSTEM CERTIFIED BY LLOYD'S REGISTER ACCORDING TO ISO 9001 : 2015, IATF 16949 : 2016, AS 9100D

The Quality Management System is Certified acc. Pressure Equipment Directive 2014/68/EU Annex 1,chapt.4.3 by TUEV and LLOYD'S
Any act of tampering, modification, alteration, counterfeiting and/or falsification and/or any other action which modifies the contents of this test certificate shall constitute a violation of applicable civil and criminal laws. Acciaierie Valbruna shall protect its rights and interests before any competent court, authority and jurisdiction.
Maxival and/or Valplus grades/products are manufactured with ladle techniques to control composition, distribution, size and shape of non-metallic inclusions for improved machinability.
The supplied product conforms to requirements expressly requested by the purchaser and conforms to requirements specified by certified norms and standards. Should the product be used for more severe, critical and/ or in any case different applications than those the material is generally intended for, any different and/or supplementary requirements shall be specifically demanded, at least, upon order of the Product by the Purchaser. Acciaierie Valbruna SpA shall not be responsible for any improper use of the products.

Vicenza, 27/02/19
VCQ012 - MEST373295
(Mod. MCE2) VSP/0703/F06301248/A060C0306/TC1A00001

Il collaudatore di stabilimento / der Werkssachverständige / Works inspector / L'agent d'usine

M.RIZZOTTO

Pagina - 3 di 3

Certificato/Certificate N°1342976Pag. 1 / 1

Data cert /Date cert.11/09/2019

d.d.t./Delivery note n.585511/09/2019

Cliente/CustomerOFFICINE OROBICHE SPA

Vs. ordine/Order.1498

Ns.rif./Int. ref.546531/07/2019

Descrizione Materiale / Material Description				
Item	Qty	Articolo / Article	Materiale / Material	Rif. Cliente / Your ref.
1	1000	Dado ASTM/ASME A/SA 194 2H UNI 5587 H=ø PG M12 ZNB 5-8µ	ASTM/ASME A/SA 194 2H	
2	500	Tirante tutto filetto ASTM/ASME A/SA 193 B7 M12x65 ZNB deidr. 5-8µ	ASTM/ASME A/SA 193 B7	8701

COLATA-ANALISI CHIMICA come da certificato di acciaieria / HEAT-CHEMICAL ANALYSIS according to mill certificate									NORME/STANDARDS
Item	Colata/Heat	C	Cr	Mn	Mo	P	S	Si	
1	156489845	0,5400		0,7400		0,0130	0,0070	0,1200	ASME II Part A ed. 2017
2	BA7559	0,4350	1,0900	0,7700	0,1800	0,0110	0,0240	0,2100	ASME II Part A ed. 2017

PROPRIETA' MECCANICHE / MECHANICAL PROPERTIES															
OTTENUTE / OBTAINED								RICHIESTE / REQUIRED							
Item	Rm	Re	A	Z	Dur/hard.		Resilienza/Impact test	Rm (N/mm²)	Re (N/mm²)	A %	Z %	Dur/hard.		Resilienza/Impact test	
	N/mm²	N/mm²	%	%	HB	°C		min	max	min	max	min	max		°C
1					272	310	20						327		20
2	949	753	23,50	61,00	283	287	20	860		720		16	50	321	20

Trattamento Termico			Dur./Hard. HB dadi/nuts				Prova allarg. per dadi		ESAME VISIVO-VIS.EXAM
Item	Tempering - °C		h	°C	>=	HB	Cone stripping on nuts		
1	Quenched and tempered macro etch examination: positive 455°		24h	540		190	Positive	Proof load positive	Positivo / Positive
2	Quenched and tempered macro etch examination: positive 593°								Positivo / Positive

Note:

Dichiariamo che il materiale è conforme alle prescrizioni dell'ordinazione e, se necessario, ai regolamenti ufficiali e alle regole tecniche corrispondenti
We state that the material is in compliance with the requirements of the order, and, if necessary, with the official regulations and related technical rules

OFFICINE OROBICHE SRL P.O.1245 TAL JOB N°418005968
N° 37 MT 222 HEAT 454322 DIMENSIONS 60,3 x 2



MANAGEMENT SYSTEM CERTIFICATION
DNV-GL
ISO 9001 - ISO 14001
QH&SAS 19001

Ita Inox S.p.A.
Strada Statale 45 bis
26010 Robecco d'Oglio (CR) - Italia

Tel + 39 0372 9801
Fax + 39 0372 921538
e-mail: sales@ita.arvedi.it
quality@ita.arvedi.it
www.arvedi.it



TEST CERTIFICATE ACCORDING TO EN 10204(2005) 3.1N°0000565548

ABNAHMEPRÜFZEUGNIS - CERTIFICAT D'ESSAIS - CERTIFICATO DI COLLAUDO

Longitudinally laser welded tubes/Laser längsnahtgeschweisste rohre/Tubes soudés longitudinalement laser/Tubi saldati longitudinalmente laser

Pag. 1di 1

Customer: T.A.L. S.P.A. TUBI ACCIAIO LOMBARDA

con socio unico

Besteller/Client/Cliente VIA G. DI VITTORIO - ZONA IND.

29017 FIORENUOLA D'ARDA PC

Customer Order N°: OF180617 -

Mill's Ita Inox N°: 0411077505 - 000010 # 0421084812 - 000020

Bestellung/Commande Client/Ordine Cliente

WerksN°/N° référence Interne/Conferma ordine

Specifications: ASTM A 312/A 312M - 16

Tolerances: ASTM A999/A999M-12

Anforderungen/Specifications/Spécifiche

Toleranzen/Tolérances/Tolleranze

Manufacturer's mark: itaInox

Marking: ASTM A312

Inspector's Stamp: A.C.

Kenzeichnung/Marquage/Marcatura

Herstellerzeichen/Marque du fabricant/ Marchio del produttore

Stempel des Sachverständigen/Poinçon de l'inspecteur/Punzone dell'ispettore

Item Pos. N°	DIMENSIONS Abmessungen Dimensions/Dimensioni	PIECES N° Stückzahl Pièces/Pezzi	METERS Meter Mètres/Metri	WEIGHT(kg) Gewicht/Poids Peso	GRADE Werkstoff/Nuance Materiale	STANDARD CODE Normbezeichnung Designation/Designazione	EXECUTION Ausführung Execution/Esecuzione
10	60.30 X 2.00 X 6000	185	1110.00	3,000.000	TP.316L 1.4404 Z3 CND 17-12-02 UNS S31603	X2 CrNiMo 17-12-2	ER

Chemical analysis acc.to: ASTM A240/ EN 10088-2/EN 10028-7 Last Edition

Steel making process :E/AOD

Schmelzanalyse/Chimique analyse/Analisi chimica

Erschmelzungsart/Procédé d'elaboration/Procedimento di elaborazione acciaio

Item N°	Manufacturer Hersteller/Fabricant/Produttore	HEAT N° Schmelze/Coulée/Colata	% C	% Si	% S	% P	% Mn	% Cr	% Ni	% Mo	% Ti	% Co	% Cu	% N
10		454322	0.026	0.340	0.0010	0.029	1.050	16.610	10.070	2.070		0.190	0.320	0.048

Mechanical test acc. to tab.: 4 - ASTM A312/A 312M

Mechanische Prüfungen/Essais mécaniques/Caratteristiche meccaniche

Item Pos. N°	HEAT N° Schmelze Coulée Colata	HOMOLOG. Zulassung Omologation Omologazione	TEST Probe Eprouvette Provino n°	SPECIMEN SIZE Abmessung Probestab Dime. Eprouvette Dimensione provetta mm.	YIELD STRENGTH Streck-Dehngrenze Limite d'élasticité Limite di snervamento 0,2% N/mm² 1%	TENSILE STRENGTH Zugfestigkeit Résistance à traction Limite di rottura N/mm²	ELONGAT. Bruchdehnung Allongement Allungamento 2"%	HARDNESS Haerte Dureté Durezza HRB
REQUIREMENT Anforderungen/Exigences/Requisiti					>=170	>= 485	>= 35.0	
10	454322		01 L	20 X 2.00	379	658	53.2	
10	454322		02 L	20 X 2.00	387	629	48.6	

Test Results

Heat treatment: 1050 °C SOLUBILIZZAZIONE / DECAPAGGIO

Technological test: Appiatt: OK / ===== / =====

Residual Corrosion Test acc.to : =====

Intergranular Corrosion Test acc.to:

Non Destructive Test acc. to: EDDY CURRENT SECONDO ASTM A999 (RIFERIMENTO DI TARATURA FORO)/E426: OK

Leak Test/Hydrostatic test to:

Uncorrect Material Test: al 100%: Favorevole

Visual and gauging control: Favorevole

Ergebnisse der Prüfungen/Résultat des essais/Risultati delle prove

Wärmebehandlung/Traitement thermique/Trattamento termico

Technologische Prüfung/Examen technologique/Prove tecnologiche

Korrosionsfördernde Rückständen/Essai residues corrosif/Prova residui corrosivi

IK Beständigkeit/Essai corr.intergr./Prova di corrosione intergranulare

Zerstörungsfreie Prüfung/Contrôle non destructif/Controllo non distruttivo

Dichtheitsprüfung/Essai d'étanchéité/Prova di tenuta

Verwechslungsprüfung/Essai P.M.I./Prova antimiscuglio

Besichtigung und Ausmessung/Contrôle visuel et dimensionel/Controllo visivo e dimensionale

Notes:

TUBI SECONDO SPECIFICA STANDARD NACE MR0175/ISO15156-MR0103 Ultima Edizione. TEST DUREZZA ROCKWELL =< 22 HRC - Materiale conforme al tipo:1.4401/316 - Composizione chimica sec. ASTM A312 Tab.1. - I.C. TEST ASTM A262 Pratica E. - TUBI SECONDO SPECIFICA STANDARD NACE MR0175/ISO15156-MR0103 Ultima Edizione. TEST DUREZZA ROCKWELL =< 22 HRC -

We certify that the delivered products comply with the specification of the order / Wir bestaetigen, dass die gelieferte Ware den Bestellvorschriften entspricht / Nous attestons que les produits livres sont conformes aux références de la commande / Noi attestiamo che il materiale spedito è conforme ai requisiti dell'ordine

Ing. Alessandro Canevari

Robecco d'Oglio, 21/11/2018

A

Mill's Inspector / Der Werksachverständige
Inspecteur de l'usine / Firma Ispettore

Arvedi





OFFICINE OROBICHE SRL P.O.1245 TAL JOB N°418005968
N° 13 MT 78 HEAT 454322 DIMENSIONS 60,3 x 2



MANAGEMENT SYSTEM CERTIFICATION
DNV-GL
ISO 9001-ISO 14001
OHSAS 18001

Ita Inox S.p.A.
Strada Statale 45 bis
26010 Robecco d'Oglio (CR) - Italia

Tel + 39 0372 9801
Fax + 39 0372 921538
e-mail: sales@ita.arvedi.it
quality@ita.arvedi.it
www.arvedi.it



TEST CERTIFICATE ACCORDING TO EN 10204(2005) 3.1N°0000567968

ABNAHMEPRÜFZEUGNIS - CERTIFICAT D'ESSAIS - CERTIFICATO DI COLLAUDO

Longitudinally laser welded tubes/Laser längsnahtgeschweisste rohre/Tubes soudés longitudinalement laser/Tubi saldati longitudinalmente laser

Pag. 1di 1

Customer: T.A.L. S.P.A. TUBI ACCIAIO LOMBARDA

con socio unico

Besteller/Client/Cliente VIA G. DI VITTORIO - ZONA IND.

29017 FIORENUOLA D'ARDA PC

Customer Order N°: OF180617 -

Mill's Ita Inox N°: 0411077505 - 000010 # 0421085504 - 000010

Bestellung/Commande Client/Ordine Cliente

WerksN°/N° référence Interne/Conferma ordine

Specifications: ASTM A 312/A 312M - 16

Tolerances: ASTM A999/A999M-12

Anforderungen/Specifications/Spécifiche

Toleranzen/Tolérances/Tolleranze

Manufacturer's mark: itaInox

Marking: ASTM A312

Inspector's Stamp: A.C.

Kenzeichnung/Markage/Marcatura

Herstellerzeichen/Marque du fabricant/ Marchio del produttore

Stempel des Sachverständigen/Poinçon de l'inspecteur/Punzone dell'ispettore

Item Pos. N°	DIMENSIONS Abmessungen Dimensions/Dimensioni	PIECES N° Stückzahl Pièces/Pezzi	METERS Meter Mètres/Metri	WEIGHT(kg) Gewicht/Poids Peso	GRADE Werkstoff/Nuance Materiale	STANDARD CODE Normbezeichnung Designation/Designazione	EXECUTION Ausführung Execution/Esecuzione
10	60.30 X 2.00 X 6000	148	888.00	2,400.000	TP.316L 1.4404 Z3 CND 17-12-02 UNS S31603	X2 CrNiMo 17-12-2	ER

Chemical analysis acc.to: ASTM A240/ EN 10088-2/EN 10028-7 Last Edition

Steel making process :E/AOD

Schmelzanalyse/Chimique analyse/Analisi chimica

Erschmelzungsart/Procédé d'elaboration/Procedimento di elaborazione acciaio

Item N°	Manufacturer Hersteller/Fabricant/Produttore	HEAT N° Schmelze/Coulée/Colata	% C	% Si	% S	% P	% Mn	% Cr	% Ni	% Mo	% Ti	% Co	% Cu	% N
10		454322	0.026	0.340	0.0010	0.029	1.050	16.610	10.070	2.070		0.190	0.320	0.048

Mechanical test acc. to tab.: 4 - ASTM A312/A 312M

Mechanische Prüfungen/Essais mécaniques/Caratteristiche meccaniche

Item Pos. N°	HEAT N° Schmelze Coulée Colata	HOMOLOG. Zulassung Omologation Omologazione	TEST Probe Eprouvette Provinò n°	SPECIMEN SIZE Abmessung Probestab Dime. Eprouvette Dimensione provetta mm.	YIELD STRENGTH Streck-Dehngrenze Limite d'élasticité Limite di snervamento 0,2% N/mm² 1%	TENSILE STRENGTH Zugfestigkeit Résistance à traction Limite di rottura N/mm²	ELONGAT. Bruchdehnung Allongement Allungamento 2"%	HARDNESS Haerte Dureté Durezza HRB
REQUIREMENT Anforderungen/Exigences/Requisiti					>=170	>= 485	>= 35.0	
10	454322		01 L	20 X 2.00	379	624	54.3	
10	454322		02 L	20 X 2.00	393	641	50.5	

Test Results

Ergebnisse der Prüfungen/Résultat des essais/Risultati delle prove

Heat treatment: 1050 °C SOLUBILIZZAZIONE / DECAPAGGIO

Wärmebehandlung/Traitement thermique/Trattamento termico

Tecnological test: Appiatt: OK / ===== / =====

Technologische Prüfung/Examen technologique/Prove tecnologiche

Residual Corrosion Test acc.to : =====

Korrosionsfördernde Rückständen/Essai residues corrosif/Prova residui corrosivi

Intergranular Corrosion Test acc.to:

IK Beständigkeit/Essai corr.intergr./Prova di corrosione intergranulare

Non Destructive Test acc. to: EDDY CURRENT SECONDO ASTM A999 (RIFERIMENTO DI TARATURA FORO)/E426: OK

Zerstörungsfreie Prüfung/Contrôle non destructif/Controllo non distruttivo

Leak Test/Hydrostatic test to:

Dichtheitsprüfung/Essai d'étanchéité/Prova di tenuta

Uncorrect Material Test: al 100%: Favorevole

Verwechslungsprüfung/Essai P.M.I./Prova antimiscuglio

Visual and gauging control: Favorevole

Besichtigung und Ausmessung/Contrôle visuel et dimensionel/Controllo visivo e dimensionale

Notes:

TUBI SECONDO SPECIFICA STANDARD NACE MR0175/ISO15156-MR0103 Ultima Edizione. TEST DUREZZA ROCKWELL =< 22 HRC - Materiale conforme al tipo:1.4401/316 - Composizione chimica sec. ASTM A312 Tab.1. - I.C. TEST ASTM A262 Pratica E. - TUBI SECONDO SPECIFICA STANDARD NACE MR0175/ISO15156-MR0103 Ultima Edizione. TEST DUREZZA ROCKWELL =< 22 HRC -

We certify that the delivered products comply with the specification of the order / Wir bestaetigen, dass die gelieferte Ware den Bestellvorschriften entspricht / Nous attestons que les produits livrés sont conformes aux références de la commande / Noi attestiamo che il materiale spedito è conforme ai requisiti dell'ordine

Ing. Alessandro Canevari

Robecco d'Oglio, 10/12/2018

A

Mill's Inspector / Der Werksachverständige
Inspecteur de l'usine / Firma Ispettore

Arvedi





COD 18516

OFFICINE OROICHE Spa
Quality Assurance

ME: 19G00422

**Acciaierie Valbruna S.p.A.**36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
Telefono 0444.968211 - Fax 0444.963836
Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
Telefono 0471.924111 - Fax 0471.924497**CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION**

In conformità a: EN 10204 (2004), 3.1 / ISO 10474 (2013), 3.1

Certificato nr. MEST423707 / 2019 /

Nach/According to/Selon

Prüfung/Test/Essai

Cliente / Besteller/Purchaser/Cliant
BREMBOMATIC PEDRALI SRL
VIA FORNASOTTO, 38/A
24040 - PONTIROLO NUOVO - BGStato di fornitura: Laminato - Solubilizzato FINITO A
Lieferzustand: FREDDO
Delivery state:
Etat de livraison:Produttore:
Hersteller/Item/Usine productrice**ACCIAIERIE VALBRUNA S.P.A.**

Ordine nr. ORD.N.19 2244

Bestell
Your order
Commande

Tipo di Elaborazione: E+AOD

Erschmelzungsart:
Melting process
Mode d'elaborationMarchi di Fabbrica:
Zeichen des Lieferwerkes
Trade marks
Sigles de l'usine productrice

Conferma ordine nr. MI19004995

Werks/Our Order/Ref. nr.

Qualità: 1.4401/1.4404/316/316L

Werkstoff/Grade/Nuance:

Punzone del Collaudatore:
Stempel des Werkstoffverständigen
Inspector's stamp/Poinçon de l'essayeur

MR

Avviso di Spedizione: A-MI19003984

Lieferanzettel/Packing list/B.L.

Marca: APLM

Markenbezeichnung
Brand / Nuance

Punzonatura: 1.4401/4/316/L

Kennzeichnung
Marking
Marquage**SPECIFICHE :**

Note:

Anforderungen / Requirements / Exigences

Aufzeichnungen / Notes / Notes

VAL STOCK 2010 1.4404/316L A,CF

AISI 316

AISI 316L

AMS 5648 M S31600 A

AMS 5653 H S31603 A,CF

ASME SA182 2017 S31600 A (0)

ASME SA182 2017 S31603 A (1)

ASME SA276 2017 S31600 A,CF (2)

ASME SA276 2017 S31603 A,CF (3)

ASME SA479 2017 S31600 A (4)

ASME SA479 2017 S31603 A (5)

ASTM A182 2018 S31600 A (6)

ASTM A182 2018 S31603 (7)

ASTM A262 2015 PRACTICE E

ASTM A276 2017 S31600 A,CF

ASTM A276 2017 S31603 A,CF

ASTM A314 2015 S31600

ASTM A314 2015 S31603

ASTM A370 2017

ASTM A479 2018 S31600 A

ASTM A479 2018 S31603 A

ASTM E10 2017

ASTM E8 2016A

EN 10088-3 2014 1.4401 A,CF

EN 10088-3 2014 1.4404 A,CF

EN 10272 2007 1.4401 A,CF

EN 10272 2007 1.4404 A,CF

ISO 148-1 2016

ISO 6506-1 2014

ISO 6892-1 2016

NACE MR0103 2015 S31600 A (8)

NACE MR0103 2015 S31603 A (9)

NACE MR0175 2015 S31600 A (A)

NACE MR0175 2015 S31603 A (B)

QQ-S-763 F 316 A,CF

QQ-S-763 F 316L A,CF

(0)Section II Part A 2017 EDITION For products machined directly from bar refer to ASME SA479.

(1)Section II Part A 2017 EDITION For products machined directly from bar refer to ASME SA479.

(2)Section II Part A 2017 EDITION

(3)Section II Part A 2017 EDITION

(4)Section II Part A 2017 EDITION

(5)Section II Part A 2017 EDITION

(6)For products machined directly from bar refer also to ASTM A479.

(7)For products machined directly from bar refer also to ASTM A479.

(8)ANSI/NACE MR0103/ISO 17945 November 23, 2015

(9)ANSI/NACE MR0103/ISO 17945 November 23, 2015

(A)ANSI/NACE MR0175/ISO 15156-3, third edition November 23,2015

(B)ANSI/NACE MR0175/ISO 15156-3, third edition November 23,2015

Tolleranza: h11 - DIN668/EN10278

Tolleranza/Allowance/Tolerance

Pos. nr. Pos. nr. Item nr. Nr. de poste	Oggetto Gegenstand Product description Description du produit	Dimensioni - mm Abmessungen Dimensions Dimension	Lunghezza - mm Länge Length Longueur	Colata Schmelze Heat Coulée	Pezzi Stückzahl Pieces Pièces	Peso - KG Gewicht Weight Poids	Lotto nr. Losnr. Lot nr. Lot nr.
0020	Quadro	25,000 x 25,000	3070/ 3070	278041		60,0	816409810

QUALITY MANAGEMENT SYSTEM CERTIFIED BY LLOYD'S REGISTER ACCORDING TO ISO 9001 : 2015, IATF 16949 : 2016, AS 9100D

Vicenza, 12/06/19

VCQ052 - MEST423707

Il collaudatore di stabilimento
der Werkstoffverständiger / Works Inspector / L'agent d'usine

M.RIZZOTTO

Direzione Qualità
Qualitätsmanagement/Quality Management/Gestion Qualité

R.BERTELLI

Direzione Prodotto
Produktmanagement/Product Management/Direction Produit

P.MESSORI

Pagina
1 di 3

**Acciaierie Valbruna S.p.A.**36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
Telefono 0444.968211 - Fax 0444.963836
Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
Telefono 0471.924111 - Fax 0471.924497**CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION**

In conformità a: EN 10204 (2004), 3.1 / ISO 10474 (2013), 3.1

Certificato nr. MEST423707 / 2019 /

Nach/According to/Selon:

Prüfung/Test/Essai:

Cliente / Besteller/Purchaser/Cliant
BREMBOMATIC PEDRALI SRL
VIA FORNASOTTO, 38/A
24040 - PONTIROLO NUOVO - BGStato di fornitura: Laminato - Solubilizzato FINITO A
Lieferzustand FREDDO
Delivery state
Etat de livraisonProduttore:
Hersteller/Hersteller/Usine productrice**ACCIAIERIE VALBRUNA S.P.A.**

Ordine nr: ORD.N.19 2244

Bestell
Your order
CommandeTipo di Elaborazione: E+AOD
Erschmelzungsart
Melting process
Mode d'elaborationMarchi di Fabbrica:
Zeichen des Lieferwerkes
Trade marks
Sigles de l'usine productrice.

Conferma ordine nr: MI19004995

Werks/Our Order/Ref nr.

Qualità: 1.4401/1.4404/316/316L
Werkstoff/Grade/NuancePunzone del Collaudatore:
Stempel des Werkstoffverständigen
Inspector's stamp/Poinçon de l'essayeur

MR

Avviso di Spedizione: A-MI19003984

Lieferanzeige/Packing list/B.L.

Marca: APML
Markenbezeichnung
Brand / NuancePunzonatura: 1.4401/4/316/L
Kennzeichnung
Marking
Marquage**TEST ALLO STATO DI FORNITURA**

Test on delivery condition Prüfung auf lieferbarem produkt test a l'etat de fourniture Prueba sobre el material así como entregado

1) L=Longitudinale/Längs, T=transversale/quer, Q=Tangenziale/tangentiell													
TEST	Provetta/ Probe/échantillon Specimen/Eprovette Larg./diam Spess. Bred. Diam. Dicke Width Diam. Thickness Larg./diam.epais mm	°C	Posiz. Saggio Position Location Emplacement 1)	Snervamento Streckgrenze Yield Stress Limite élastique Rp 0,2% N/mm2	Snervamento Streckgrenze Yield Stress Limite élastique Rp 1% N/mm2	Resistenza Zugfestigkeit Tensile strength Résistance à traction Rm N/mm2	Allungamento Bruchdehnung Elongation Allongement A5 % E 4d %	Strizione Einschnürung Reduction of area Striction Z % RA %	Resilienza Kerbschlagarbeit Impact Value Resilience KV J	Durezza Härte Hardness Dureté HB			
Valori richiesti		min max		207	235	517 900	20 30	- 50	100	140 235			
Anforderungen/Required values Valeurs demandées													
A	10	20	L	456	518	645	48 50	66 66	201 198 201 214				

TEST ALLO STATO DI FORNITURA

Test on delivery condition / Prüfung auf lieferbarem produkt / Test a l'etat de fourniture / Prueba sobre el material así como entregado

TEST	min	max
A Dimensioni grano x ASTM E112		5

Mechanical properties according to ASTM A370.

Brinell hardness according to ASTM E10

Tensile testing according to ASTM E8

Impact testing according to ISO 148-1

Brinell hardness according to ISO 6506-1

Tensile testing according to ISO 6892-1

Analisi chimica

Chemische Zusammensetzung/Chemical Analysis/Analyse chimique

Colata /Heat Schmelze/Coulée	min max	0,030	1,00	1,25 2,00	16,50 18,00	2,00 2,50	10,00 13,00	0,040	0,030	0,100	-	-	-	-	-
	C %	Si %	Mn %	Cr %	Mo %	Cu %	Ni %	P %	S %	N %					
278041	0,015	0,43	1,37	16,70	2,00	0,46	10,02	0,030	0,027	0,078					

Corrosion test in 16% sulfuric acid and copper sulfate solutions

Test standard: ASTM A262-Practice E

Test	Heat treatment before test	Length of Period (h)	Test temp (°C)	Bend Angle (°)	Ø spindle (mm)	Result of visual inspection at 20 x magnification after bend test	Result
Intergranular corrosion	Sensitization	15	boil	180	5	Absence of cracks	SATISFACTORY

QUALITY MANAGEMENT SYSTEM CERTIFIED BY LLOYD'S REGISTER ACCORDING TO ISO 9001 : 2015, IATF 16949 : 2016, AS 9100D

Vicenza, 12/06/19

VCQ052 - MEST423707

Il collaudatore di stabilimento
der Werkstoffverständige / Works Inspector / L'agent d'usine

M.RIZZOTTO

Direzione Qualità

Qualitätsmanagement/Quality Management/Gestion Qualité

R.BERTELLI

Direzione Prodotto

Produktmanagement/Product Management/Direction Produit

P.MESSORI

Pagina
2 di 3



Acciaierie Valbruna S.p.A.

36100 VICENZA (Italia) - Viale della scienza, 25 z.i.
 Telefono 0444.968211 - Fax 0444.963836
 Stab.: 39100 BOLZANO (Italia) - Via A. Volta, 4/37
 Telefono 0471.924111 - Fax 0471.924497

CERTIFICATO DI COLLAUDO - ABNAHMEPRUEFZEUGNIS - INSPECTION CERTIFICATE - CERTIFICAT DE RECEPTION

In conformità a: EN 10204 (2004), 3.1 / ISO 10474 (2013), 3.1
 Nach/According to/Selon:

Certificato nr. MEST423707 / 2019 /
 Prüfung/Test/Essai

Cliente / Besteller/Purchaser/Cliant
 BREMBOMATIC PEDRALI SRL
 VIA FORNASOTTO, 38/A
 24040 - PONTIROLO NUOVO - BG

Stato di fornitura: Laminato - Solubilizzato FINITO A FREDDO
 Lieferzustand: Delivery state: Etat de livraison:

Produttore:
 Hersteller/Item/Usine productrice

ACCIAIERIE VALBRUNA S.P.A.

Ordine nr. ORD.N.19 2244
 Bestell:
 Your order
 Commande

Tipo di Elaborazione: E+AOD
 Erschmelzungsart:
 Melting process:
 Mode d'elaboration:

Marchi di Fabbrica:
 Zeichen des Lieferwerkes
 Trade marks
 Sigles de l'usine productrice



Conferma ordine nr: MI19004995
 Werks/Our Order/Réf nr.

Qualità: 1.4401/1.4404/316/316L
 Werkstoff/Grade/Nuance:

Punzone del Collaudatore:
 Stempel des Werkssachverständigen
 Inspector's stamp/Pointon de l'essayeur

MR

Avviso di Spedizione: A-MI19003984
 Lieferanzeige/Packing list/B.L.

Marca: APM L
 Markenbezeichnung:
 Brand / Nuance:

Punzonatura: 1.4401/4/316/L
 Kennzeichnung:
 Marking:
 Marquage:

Corrosion test in 16% sulfuric acid and copper sulfate solutions

Test standard: UNI EN ISO 3651-2 Method A

Test	Heat treatment before test	Length of Period (h)	Test temp (°C)	Bend Angle (°)	Ø spindle (mm)	Result of visual inspection at 20 x magnification after bend test	Result
Intergranular corrosion	Sensitized T1	20	boil	90	5	Absence of cracks	SATISFACTORY

Reduction ratio = 47,9 : 1
 Solution annealed 1070°C for 0.5h/ water

Sono state soddisfatte tutte le condizioni richieste
 Die gestellten Anforderungen sind erfüllt
 The material has been furnished in accordance with the requirements
 Le matériel a été trouvé conforme aux exigences

Controllo antimescolanza: OK
 Verwechslungsprüfung: spectralanalytisch durchgeführt
 Antimixing testing performed: OK
 Contrôle antimélange: satisfaisant

Controllo visivo e dimensionale: soddisfa le esigenze
 Beschichtigung und Ausmessung: ohne Beanstandung
 Visual inspection and dimensional checks: satisfactory
 Contrôle visuel et dimensions: satisfaisant

Melted and manufactured in Italy No welding or weld repair Material free from Mercury contamination

We declare that the finished product is checked for radioactive contamination through Portal System when it leaves the production plant.

The Quality Management System is also Certified according Pressure Equipment Directive 2014/68/EU Annex 1, chapt.4.3 by TÜV and LLOYD'S
 Any act of tampering, modification, alteration, counterfeiting and/or falsification and/or any other action which modifies the contents of this test certificate shall constitute a violation of applicable civil and criminal laws. Acciaierie Valbruna shall protect its rights and interests before any competent court, authority and jurisdiction.

Maximal and/or Valplus grades/products are manufactured with ladle techniques to control composition, distribution, size and shape of non-metallic inclusions for improved machinability.

The supplied product conforms to requirements expressly requested by the purchaser and conforms to requirements specified by certified norms and standards. Should the product be used for more severe, critical and/or in any case different applications than those the material is generally intended for, any different and/or supplementary requirements shall be specifically demanded, at least, upon order of the Product by the Purchaser. Acciaierie Valbruna SpA shall not be responsible for any improper use of the Products.

VSP013HEPT123174287624F47615TE45AC40

QUALITY MANAGEMENT SYSTEM CERTIFIED BY LLOYD'S REGISTER ACCORDING TO ISO 9001 : 2015, IATF 16949 : 2016, AS 9100D

Vicenza, 12/06/19 VCQ052 - MEST423707	Il collaudatore di stabilimento der Werkssachverständige / Works Inspector / L'agent d'usine M. RIZZOTTO	Direzione Qualità Qualitätsmanagement/Quality Management/Gestion Qualité R. BERTELLI	Direzione Prodotto Produktmanagement/Product Management/Direction Produit P. MESSORI	Pagina 3 di 3
--	--	--	--	------------------

Customer: OFFICINE OROBICHE S.r.l.
 Order : 1166 - 13.06.2019 Project: Our ref.: OCVEIT201900003162
 Descr.: 33.40x2.77 W
 Heat n° or Pcs. marking : VBK123304 Qta/Qty:60 Posiz./Item n.: 10
 Protocol : CTCERC201900006200 * CERTIFIED TRUE COPY * Issued 17/06/19



INSPECTION CERTIFICATE

No.270,Sec.4,Jungshan Rd.,Shijou Shiang,
 Chang-Hwa,Taiwan,R.O.C
 http://www.ycinox.com
 TEL : +886-4-8899666 FAX : +886-4-8899770



Customer		RACCORTUBI S.P.A		Customer No.		FIT016		Certificate No.		190306P0039										
Product		STAINLESS STEEL WELDED PIPE		Commercial Invoice No.		J19020170		Date of Issue		2019/03/05										
Specification		ASTM A312/A312M-17		PO No.		OS000269 S		Order No.		EP18120282-3										
Steel Grade		TP316/316L		Finish		PICKLED		P.I.No.		PI 1812-162										
Item No.	Heat No.	Product Description		Chemical Composition(%)								Tension Test			*01 HR	*02 HT	*03 HY	*04 ET	*05 F1	Coil No
		Dimension	Quantity Pcs	C	Si	Mn	P	S	Ni	Cr	Mo	0.2% Y.S.	T.S.	E.L.						
		Specification	Min. Max.	X10 ⁻³	X10 ⁻²	X10 ⁻²	X10 ⁻³	X10 ⁻³	X10 ⁻²	X10 ⁻²	X10 ⁻²	MPa	%	HRB	°C	MPa				
1	VBK123304	1"(33.4mm) x SCH-10S(2.77mm) x 6M	84	35	100	200	45	30	1000	1600	200	205	515	35		1040				
		TOTAL:	84	16	55	113	35	2	1012	1665	203	273	583	54	74	1060	17.0	OK	OK	7CS57410
Remark				Note																
*STANDARDS:ASTM A312/ASME SA312 (2017 Section II Part A) *Weld Factor = 0.8 (Acc. to ASME B31.3 Table 302.3.4)				100% Alloy Verification(PMI) : OK Intergranular Corrosion Test (according to ASTM A262 Practice E) : OK Hardness according to NACE MR0175 / ISO 15156-3:2015 : Satisfactory. Hardness according to NACE MR0103-2015 : Satisfactory. Visual Inspection(Surface) and Dimension Check : OK																
Surveyor By				We hereby certify that material described herein has been manufactured and tested with satisfactory results in accordance with the requirement of the above material specification. The material described above has been found free of radiation by raw material supplier.																
N/A				1.ISO 9001:2015 (No.01 100 056299) 2.PED / AD2000 (No.01 202 TWN/Q-05 0105) 3.ISO 14001:2004 (No.01 104 822 096385) 4.OHSAS 18001:2007 (No.01 113 822 096385)																
				Manager of Quality Assurance Department 																

DUEVI

FORNITURE INDUSTRIALI SRL

DUEVI SRL
Via Albert Sabin, 15 20873 - CAVENAGO BRIANZA (MB)
Tel. +39 02 95335174 Fax +39 02 95335175
Registro imp. Monza C.F./Partita IVA/VAT no. IT 06473230966
E-mail info@duevifornitureindustriali.it Web www.duevifornitureindustriali.it

SISTEMA QUALITA'
CERTIFICATO ISO 9001:2015
RINA NR. 7026/02/S

INSPECTI ON CERTI FI CATE EN 10204 : 2004 / 3.1										Nr. / No. 001730		Data / Dated 17/06/2019							
OFFI CI NE OROBI CHE SRL VIA SERENA, 10 24010 PONTERANICA - BG IT										DDT / Delivery note 1706 Packing List Pag. 1 Vs.Ordine nr./Yr. P.O. no. MB VS ORD. NR. 970 DEL 21/05/2019 Posizione / Item 002 Ns. rif. / Our ref.									
Codice colata Heat Code		Nr. colata Heat No.		Origine Origin of Steel				Quantita' Quantity		Descrizione Description									
		561540		VALBRUNA				198		TAPPI MASCHIO T.E. F316L 1/2" NPT 3000									
Elementi / Elements		C	Si	Mn	S	P	Cr	Ni	Mo	Ti	Cu	V	Nb	N	Al				
LADLE ANALYSIS		0,022	0,430	1,650	0,026	0,029	17,570	12,540	2,090					0,044					
		CE LF																	
		5,065										*****							
		CE LF= C+ Mn/6+ (Cr+ Mo+ V)/5+ (Ni+ Cu)/15										CE SF= C+ Mn/6		F1= Cu+ Ni+ Cr+ Mo+ V		F2= Cr+ Mo		PREN	
DUREZZA						Snervamento> 0,2%		Snervamento> 1,0%		Rottura		Allungamento		Contrazione					
HARDNESS						Yeld Strength> 0,2%		Yeld Strength> 1,0%		Tensile		Elongation		Reduction of area					
HBW						MPa		MPa		MPa		%		%					
223						503,0				645,0		42,0		67,0					

Provetta Test specimen		Forma Shape		°C		RESI LI ENZA / I MPA CT TEST						
Sez/ Sect mm2	L. mm	1= ☐	2= ☐			Tipo/ Type	Provetta/ Test Specimen	°C	1-Joule	2-Joule	3-Joule	Media/ Average
									0,00	0,00	0,00	

Materiale in accordo a: Material according to:		☒	ASTM/ASME CODE SECT. II, PART. A, 2017 EDITION ASTM/ASME A/SA182 F316/316L									
		☒	NACE MR-0175/2009 ISO 15156									
		☒	NACE MR-0103/2012									
		☒	Q.A.S.IN ACCORD.WITH PRESSURE EQUIPMENT DIRECT. 2014/68/UE(PED) ANNEX i, PARAGRAPH 4.3									

Tratt.Term./ Heat treatment		(1)	(2)	(3)	(4)
		B	D	X	X

Dimensioni in accordo a: Dimensions according to:		004		ASTM/ASME B 16.11 ED. 2013	
--	--	-----	--	----------------------------	--

Marcatura in accordo a: Marking according to:		004		Vis. & Dim.	
--	--	-----	--	-------------	--

Note / Notes

Conforme al certificato originale rilasciato dall'acciaieria o forgiatore e relativo laboratorio di analisi e prove.
According to original certification issued by steel plant and relative analysis and tests laboratory.

(1) PRODUZIONE ACCIAIO
Steel Production

A - CALMATO ELABORATO AL FORNO ELETTRICO
Killed steel made by electric furnace

B - ELABORATO AL FORNO ELETTRICO
Made by electric furnace

(2) TRATTAMENTO TERMICO
Heat treatment

A - NORMALIZZATO
Normalized

B - NORMALIZZATO E RINVENUTO
Normalized and tempered

C - BONIFICATO
Quenched and tempered

D - SOLUBILIZZATO
Solution treatment and quenched

(3)

X - CERTIFICATO SECONDO EN 10204/3.1
Certificate according to EN 10204/3.1

(4)

X - CONTROLLO VISIVO E DIMENSIONALE SODDISFACENTE
Satisfactory visual and dimensional test

UFFI CI O CONTROLLO QUALI TA' QUALI TY CONTROLDEPARTMENT DUEVI s.r.l.	UFFI CI O CONTROLLO QUALI TA' I NSPECTI ON AUTORI THY	MARCHI O PRODUZI ONE MANUFACTURER'S SYMBOL
--	--	---

INSPECTION CERTIFICATE EN 10204 : 2004 / 3.1	Nr. / No. 001596	Data / Dated 04/06/2019
--	------------------	-------------------------

OFFICINE OROBICHE SRL

VIA SERENA, 10
24010 PONTERANICA - BG
IT

DDT / Delivery note 1582
Packing List
Pag. 2
Vs.Ordine nr./Yr. P.O. no. MB VS ORD N 1015 DEL 27.05.2019
Posizione / Item 004
Ns. rif. / Our ref.

Codice colata Heat Code	Nr. colata Heat No.	Origine Origin of Steel	Quantita' Quantity	Descrizione Description
572	572356	COGNE	100	NIPPLI ESAGONALI F316L 1/2" MM. NPT 3000

Elementi / Elements	C	Si	Mn	S	P	Cr	Ni	Mo	Ti	Cu	V	Nb	N	Al
LADLE ANALYSIS	0,022	0,350	1,790	0,026	0,032	17,090	10,200	2,070		0,340			0,074	

CE LF
4,855

CE LF= C+ Mn/6+ (Cr+ Mo+ V)/5+ (Ni+ Cu)/15 CE SF= C+ Mn/6 F1= Cu+ Ni+ Cr+ Mo+ V F2= Cr+ Mo PREN

DUREZZA HARDNESS

HBW
180

Snervamento > 0,2%
Yeld Strength > 0,2%
MPa

343,0

Snervamento > 1,0%
Yeld Strength > 1,0%
MPa

Rottura
Tensile
MPa

606,0

Allungamento
Elongation
%

55,0

Contrazione
Reduction of area
%

75,9

Provetta Test specimen		Forma Shape	°C	RESILIENZA / IMPACT TEST						
Sez/ Sect mm2	L. mm	1= - 2=		Tipo/ Type	Provetta/ Test Specimen	°C	1-Joule	2-Joule	3-Joule	Media/ Average
							0,00	0,00	0,00	0,00

Materiale in accordo a: ☒ ASTM/ASME CODE SECT. II, PART. A, 2017 EDITION ASTM/ASME A/SA182 F316/316L
Material according to: ☒ NACE MR-0175/2009 ISO 15156
☒ NACE MR-0103/2012
☒ Q.A.S.IN ACCORD.WITH PRESSURE EQUIPMENT DIRECT. 2014/68/UE(PED) ANNEX i, PARAGRAPH 4.3

Tratt.Term./ Heat treatment	(1) (2) (3) (4)
B D X X	

Dimensioni in accordo a: Dimensions according to:	004	ASTM/ASME B 16.11 ED. 2013
--	-----	----------------------------

Marcatura in accordo a: Marking according to:	004	Vis. & Dim.	
--	-----	-------------	--

Note / Notes

Conforme al certificato originale rilasciato dall'acciaieria o forgiatore e relativo laboratorio di analisi e prove.
According to original certification issued by steel plant and relative analysis and tests laboratory.

- | | | |
|--|---|--|
| (1) PRODUZIONE ACCIAIO
Steel Production | A - CALMATO ELABORATO AL FORNO ELETTRICO
Killed steel made by electric furnace | B - ELABORATO AL FORNO ELETTRICO
Made by electric furnace |
| (2) TRATTAMENTO TERMICO
Heat treatment | A - NORMALIZZATO
Normalized | B - NORMALIZZATO E RINVENUTO
Normalized and tempered |
| (3) | X - CERTIFICATO SECONDO EN 10204/3.1
Certificate according to EN 10204/3.1 | C - BONIFICATO
Quenched and tempered |
| (4) | X - CONTROLLO VISIVO E DIMENSIONALE SODDISFACENTE
Satisfactory visual and dimensional test | D - SOLUBILIZZATO
Solution treatment and quenched |

UFFICIO CONTROLLO QUALITA' QUALITY CONTROL DEPARTMENT DUEVI s.r.l.	UFFICIO CONTROLLO QUALITA' INSPECTION AUTHORITY	MARCHIO PRODUZIONE MANUFACTURER'S SYMBOL
---	--	---

Bolla/N°BL/DL N°/Lic.	N° Certificat/Paul N°	Producteur/Usine product./Manufact./Hersteller.
129824	9K0000569	080040 AP EUROPE BE

RAPPORTO DI PROVA

EN 10204:2005 2.2

Prodotto Erzeugnisform Produit Product		Nast EN10088:2 1.4404 (316L) 2B 0,60 X 3000,0										
Cliente e/o destinatario - Client et/ou destinataire - Besteller und/oder Empfänger Purchaser and/or Consignee		N° ordine cliente - Votre référence - Ihre Auftrags Nr - Your reference ORD. US 9032 04/03/19										
Qualità e specifiche - Niveau et spécifications techniques - Leistungs- und Prüfbedingungen - Quality and Specifications 1.4404 (316L)												
Stato di fornitura - Etat de livraison - Lieferzustand - As delivered(1) 2B		Stato metalurgico - Etat métallurgique - Heat treatment condition - Lieferzustand										
N° Confirma d'ordine Accusé de réception Auftragsbestätigung N° Acknowledgement N°	Posizione N° Poste Post. Nr Item N°	N° lotto interno N° de lot interne Internal Lot N°	Profilo Prüfl. Profile Sharps									
903132350	12	0900147 0900148	Nast EN10088:2 0,6000 1000,000									
			Peso netto Masse Gewicht Weight 8520,000 8350,000									
N° Colore N° de couleur Schmelze Nr Heat N°	Lotto fornitore Lot Fournisseur Band Nr Master batch	Trazione - Traction - Zugversuch - Tensile test				Resilienza - Résilience - Kerbschlagzähigkeit - Notch Toughness				Media Moyenne Mittelwerte Average	Durezza Dureté Härte Hardness	
		Lim. di snervamento Limite d'élasticité Streckgrenze Yield Strength		Carico di rottura Résistance traci. Zugfestigkeit Tensile strength	Allungamento Élongation Bruchdehnung Elongation %	Allungamento Élongation Bruchdehnung Elongation %	Resilienza Résilience Kerbschlagzähigkeit Notch toughness J _{KV} %	Forma Type Form Type	Sens Sens Richt. Direct.			Valori Valeurs individuelles Einzelwerte Individual values
		0.2% RP Mpa 0	1% RP1 Mpa 0									
841093	84109340	312	328	657	46						127	
841093	84109340	312	328	657	46						127	
		C	SI	MN	NI	CR	MO		N	S	P	
841093	84109340	0,018	0,500	1,320	10,070	16,600	2,320		0,036	0,001	0,036	
841093	84109340	0,018	0,500	1,320	10,070	16,600	2,320		0,036	0,001	0,036	
Test : (I = OK)												
841093												
841093												
C.INT = Controllo integrità SCHW. = Schweißnahtprüfung		EDD.C = Controllo non distruttivo PLANO = Flugsplattentest		P.P.N. = Prova pneumatica REVVL = Applik. saldatura (Re		C.MES = Controlli meccanici F.RET = Bassi di riferimento		Note - Notes - Remarques - Bemerkungen				
Il 19/03/19 Firma del responsabile												

199888



QA - System
DIN ISO 9001 - 2015
Certificate Nr. DE16/819942393.00



Material / Werkstoff	A182/A182M-18 // SA182/SA182M-17: F316 / F316L	Certificate No.	393955-48
Requirements / Anforderungen	A182/A182M-18 // SA182/SA182M-17 // NACE MR0175 // ISO 15156-3-15 // MR0103-15 // ASME Code Sect.II Part A-17 // ASME/ANSI B16.5-17	Date of issue.	21.02.2019
Heat treatment / Wärmebehandlung	Solution annealed at 1050°C, rapid cooling by quenching in cold water	Purchase Order No.	MSA18-50239
		Works No.	2058627

Line Item No. / Pos	Qty / Menge	Description / Gegenstand	Specimen No. / Probe	Heat No. / Schmelz Nr.
48	200	SW-FI ASME B16.5 NPS 1" Cl.150 Form Facing:RF Sched.40s	C1368	FF6672

Chemical analysis / Chemische Analyse							
	C%	Si%	Mn%	P%	S%	Mo%	Cr%
min	0	0	0	0	0	2	16
max	0,03	1	2	0,045	0,03	3	18
[HA]	0,021	0,31	1,66	0,042	0,030	2,04	16,87
[PA]	-	-	-	-	-	-	-
	Ni%	Cu%	Ti%	W%	Co%	Nb%	N%
min	10	-	-	-	-	-	0
max	14	-	-	-	-	-	0,1
[HA]	10,04	-	-	-	-	-	0,059
[PA]	-	-	-	-	-	-	-

Mechanical properties / Mechanische Eigenschaften						
Specimen No. Probe Nr.	Temperature Temperatur	Yield Strength Dehngrenze	Tensile Strength Zugversuch	Elongation Dehnungsgrenze	Reduction of Area Einschnürung	Direction Ausrichtung
-	[°C]	Rp 0,2 [N/mm²]	Rp 1,0 [N/mm²]	[N/mm²]	4D [%]	[%]
		>205	-	>515	>30	>50
C1368	RT	258	301	590	58	72
						transversal

Impact Strength / Kerbschlagzähigkeit					
Specimen No. Probe Nr.	Temperature Temperatur	Impact energy Schlagarbeit	Lateral expansion Laterale Breitung	Hardness Härte	Direction Ausrichtung
-	[°C]	[J]	[mm]	[HBW]	-
C1368	RT	-	-	170 - 175	transversal

Additional remarks / Zusätzliche Bemerkungen	
Intergranular corrosions test: satisfactory to ASTM - A262 Practice E	

Surface and dimensional check according to ASME B16.5
100% PMI with X-Ray Fluorescence (XRF) Spectro xSORT & found satisfactory
Mechanical tests according to ASTM A370
Steel made by electric furnace
[HA] = Ladle Analysis / [PA] = Product Analysis
Flange facing finish according to ASME B16.5 - 2017 Cl.6.4.5
Melting Process = EAF

Workinspector - Der Werkssachverständige

Patrick Oberhauser

 Correspondentieadres: Aperam Genk Svinnenvijerweg 5, Poort Genk 7523 3600 Genk, Belgium Tel. +32 (0)89 30 21 11		A04 MILL CERTIFICATE BS EN 10204/3.1 CERTIFICAT DE RECEPTION NF EN 10204/3.1 ABNAHMEPRÜFZEUGNIS DIN EN 10204/3.1		N-Nr-N 17K0007605-CE V01																																																																					
		A02 Factory Production Control certified by TUV SUD Industrie Service GmbH with certificate nr 0036-CPD-43-2011. In compliance with the Construction Product Directive nr 89/106/EEC.		OFFICINE OROBICHE SpA Quality Assurance ME 13401 112																																																																					
A01 Manufacturer's works order number N° de la commande usine productrice Werksauftragsnummer 80367672/01-20793/660/01		A08 Surveyor's mark Cachet de l'expert Stempel des Werkssachverständigen 		Z05 Purchaser and/or consignee Client et/ou destinataire Besteller und/oder Empfänger																																																																					
A06 Product - Produit - Erzeugnis COIL, COLD ROLLED, UNTRIMMED, FINISH 2B COIL, LAMINE A FROID, BORDS NON REFENDUS, FINI 2B CI, KALTGERW., GEGLÜHT+GEBÜHT, LECHT NACHGEW., UNBESAMT		B01 Steel designation Désignation de l'acier Stahlbezeichnung EN 10088 4 2009 1.4404/1.4401		A07 Purchaser's order number N° de commande client Kundenbestellnummer																																																																					
B02 Finish Présentation Ausführung 2B		C70 Steelmaking process Mode d'élaboration de l'acier - Stahlherstellungsverfahren Prod. proces: Electric arc furnace - VOD/AOD - Continuous casting Proc. fabric.: Four à arc - VOD/AOD - Coulée continue Fertigungsablauf: Elektro-Ofen - VOD/AOD - Stranggussanlage		A09 Customer article number N° article client Artikelnummer des Kunden																																																																					
B03 Any supplementary requirements Prescriptions supplémentaires - Zusätzliche Anforderungen		C70 Product delivery condition Etat de livraison du produit - Lieferzustand Solution treated: Hypertempe: 1050 °C Loesungsgegl+abgeschreckt: Forced air-water/air forcé-eau Gebläse Luft-Wasser		B04																																																																					
Z08																																																																									
B07 Identification of the product Identification du produit - Identifizierung des Erzeugnisses MELTED IN BELGIUM, MADE IN BELGIUM		Dimensions Dimensions - Abmessungen Thickness Épaisseur - Stärke 0.57 mm		B08 Number of pieces Nb de pièces - Stückzahl 1																																																																					
Coll n. N. Bobine - Band Nr. 64903952		Heat n. N. Coulée - Schmelz Nr. 649039		B13 Net weight Poids net - netto Gewicht 18400 KG																																																																					
CHEMICAL ANALYSIS - ANALYSE CHIMIQUE - CHEMISCHE ZUSAMMENSETZUNG																																																																									
<table border="1"> <thead> <tr> <th></th> <th>C</th> <th>Si</th> <th>Mn</th> <th>Ni</th> <th>Cr</th> <th>Mo</th> <th>Ti</th> <th>N</th> <th>S</th> <th>P</th> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> </tr> </thead> <tbody> <tr> <td>Required - Exigé %mini Anforderung. %maxi</td> <td>0.030</td> <td>1.00</td> <td>2.00</td> <td>10.00</td> <td>16.50</td> <td>2.00</td> <td></td> <td>0.100</td> <td>0.015</td> <td>0.045</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Cast Analysis Analyse coulée Analyse Schmelze</td> <td>0.021</td> <td>0.46</td> <td>1.27</td> <td>10.02</td> <td>16.55</td> <td>2.02</td> <td></td> <td>0.036</td> <td>0.001</td> <td>0.031</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td>C71</td> <td>C72</td> <td>C73</td> <td>C74</td> <td>C75</td> <td>C76</td> <td>C77</td> <td>C78</td> <td>C79</td> <td>C80</td> <td>C81</td> <td>C82</td> <td>C83</td> <td>C84</td> <td>C85</td> <td>C86</td> </tr> </tbody> </table>							C	Si	Mn	Ni	Cr	Mo	Ti	N	S	P							Required - Exigé %mini Anforderung. %maxi	0.030	1.00	2.00	10.00	16.50	2.00		0.100	0.015	0.045							Cast Analysis Analyse coulée Analyse Schmelze	0.021	0.46	1.27	10.02	16.55	2.02		0.036	0.001	0.031								C71	C72	C73	C74	C75	C76	C77	C78	C79	C80	C81	C82	C83	C84	C85	C86
	C	Si	Mn	Ni	Cr	Mo	Ti	N	S	P																																																															
Required - Exigé %mini Anforderung. %maxi	0.030	1.00	2.00	10.00	16.50	2.00		0.100	0.015	0.045																																																															
Cast Analysis Analyse coulée Analyse Schmelze	0.021	0.46	1.27	10.02	16.55	2.02		0.036	0.001	0.031																																																															
	C71	C72	C73	C74	C75	C76	C77	C78	C79	C80	C81	C82	C83	C84	C85	C86																																																									
Positive material identification carried out: OK Tests de vérification de la conformité de la nuance fournie: OK Ververechslungsprüfung wurde durchgeführt: OK C04																																																																									
Location (1) MECHANICAL PROPERTIES - PROPRIETES MECANQUES - MECHANISCHE WERTE EN ISO 6892-1 B																																																																									
Room temperature - Température ambiante - Raumtemperatur																																																																									
Test temperature (°C):																																																																									
<table border="1"> <thead> <tr> <th rowspan="2">Direction (2)</th> <th rowspan="2">Yield or proof strength Limite d'élasticité Dehngrenze MPa</th> <th rowspan="2">Tensile Strength Résistance à la traction Zugfestigkeit MPa</th> <th rowspan="2">Elongation after fracture Allongement après rupt. Bruchdehnung %</th> <th rowspan="2">Hardness Dureté Hlaerte</th> <th rowspan="2">Yield or proof strength Limite d'élasticité Dehngrenze MPa</th> <th rowspan="2">Tensile str. Résist. MPa Zugfestigkeit</th> <th rowspan="2">Elongation % Allongement Bruchdehnung</th> </tr> <tr> <th>Required Exigé Anforderung</th> <th>Obtained Obtenu Ergebnisse</th> </tr> </thead> <tbody> <tr> <td rowspan="2">1</td> <td>Rp0.2%</td> <td>Rp1%</td> <td>Rm</td> <td>50um</td> <td>80mm</td> <td>HRBW</td> <td>Rp0.2%</td> </tr> <tr> <td>240</td> <td>270</td> <td>530</td> <td>40</td> <td>40</td> <td></td> <td></td> </tr> <tr> <td rowspan="2">T</td> <td>307</td> <td>318</td> <td>651</td> <td>51</td> <td>48</td> <td>77</td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>						Direction (2)	Yield or proof strength Limite d'élasticité Dehngrenze MPa	Tensile Strength Résistance à la traction Zugfestigkeit MPa	Elongation after fracture Allongement après rupt. Bruchdehnung %	Hardness Dureté Hlaerte	Yield or proof strength Limite d'élasticité Dehngrenze MPa	Tensile str. Résist. MPa Zugfestigkeit	Elongation % Allongement Bruchdehnung	Required Exigé Anforderung	Obtained Obtenu Ergebnisse	1	Rp0.2%	Rp1%	Rm	50um	80mm	HRBW	Rp0.2%	240	270	530	40	40			T	307	318	651	51	48	77																																				
Direction (2)	Yield or proof strength Limite d'élasticité Dehngrenze MPa	Tensile Strength Résistance à la traction Zugfestigkeit MPa	Elongation after fracture Allongement après rupt. Bruchdehnung %	Hardness Dureté Hlaerte	Yield or proof strength Limite d'élasticité Dehngrenze MPa									Tensile str. Résist. MPa Zugfestigkeit	Elongation % Allongement Bruchdehnung																																																										
						Required Exigé Anforderung	Obtained Obtenu Ergebnisse																																																																		
1	Rp0.2%	Rp1%	Rm	50um	80mm	HRBW	Rp0.2%																																																																		
	240	270	530	40	40																																																																				
T	307	318	651	51	48	77																																																																			
Impact strength test Essai de résilience Kerbschlagzähigkeitstest C40 (°C) C44																																																																									
Corrosion test Test de corrosion Korrosionstest EN ISO 3651/2 - A:OK D51																																																																									
Internal cleanliness: A: B: C: D:																																																																									
Location of the sample (1) Emplacement de l'échantillon Lage des Probenabschnittes 1. Front - Début - Anfang 2. Back - Fin - Ende 3. Middle - Milieu - Mitte C01																																																																									
The delivery is in accordance with the order La fourniture est conforme aux exigences de la commande Die Lieferung entspricht den Bestellbedingungen Z01																																																																									
Packaging list Avis d'expédition Liefererscheinnummer 2017029510-100076 A10																																																																									
Organisation inspection Organisme et/ou service contrôle Überwachungsabteilung A06																																																																									
Quality Department 14/2/2017 The inspector Le responsable Der Werkssachverständige D. Raemaekers Z02																																																																									
Direction of the test pieces (2) Orientation des éprouvettes Probenrichtung T. Transverse - Transvers - Quer L. Longitudinal - Long - Längs C02																																																																									

 Correspondance address Adresse de correspondance - Adresse für briefwechsel Swinnenwijerweg 5, Poort Genk 7523 3600 Genk, Belgium	Annex to certificate 17K0007605-01 V01 Annexe du CCPU Anlage zum Zeugnis		Certificate CCPU - Zeugnis 17K0007605-CE V01
	Certificate of Production Control Number Numéro du certificat de contrôle de la production Zeugnisnummer von Produktionskontrolle 0036-CPD-43-2011		
	Year 11 Année Jahr		
Manufacturer's works order number N° de la commande usine productrice Werksauftragsnummer 80367672/01-20793/660/01	Purchaser and/or consignee Client et/ou destinataire Besteller und/oder Empfänger		Purchaser's order number N° de commande client Kundenbestellnummer Customer article number N° d'article client Artikelnummer des Kunden
Identification of the product Identification du produit - Identifizierung des Erzeugnisses	Dimensions Dimensions - Abmessungen		
Coil n° N° de bobine - Band Nr 64903952	Thickness Épaisseur - Dicke 0.57 mm	Width Largeur - Breite 1013.00 mm	Length Longueur - Länge
Stainless steel / Acier inoxydable / Rostfreier Stahl			
EN10088-4		OFFICINE OROBICHE SpA Quality Assurance ME 13601	
Intended uses : building constructions or civil engineering Usages prévus : construction immobilière ou génie civil Vorgesehene Verwendungen : Hochbauten und Ingenieurbauwerke			
Declaration of performance : Déclaration des performances : Leistungserklärung : DoP available on website Aperam : www.aperam.com/europe/news-publications/documentation/certifications/approvals DOP.Nr.GNK.08.01			
Steel Acier 1.4404/1.4401 Werkstoff			
Cold rolled Laminé à froid Kaltgewalzt			
Coil / bobine / Band			
Elongation / Allongement / Bruchdehnung Tensile strength / Résistance à la traction / Zugfestigkeit Yield strength / Limite d'élasticité / Dehngrenze Impact strength / Résistance au choc / Kerbschlagzähigkeit Weldability / Aptitude au soudage / Schweißseignung Durability / Durabilité / Dauerhaftigkeit Characteristics expressed as indicated in the above mentioned DOP Tolerances on dimension and shape / Tolerances sur les dimensions et sur forme / Grenzabmasse und Formtoleranzen : EN ISO 9445			
Regulated substance : no performance determined Substance réglementée : aucune performance déterminée Regulierter Stoff: keine Leistung festgestellt			